
Statistic Preservation Does Not Identify Attack-Suppression Preservation: Limits of Composed Defenses in Collaborative Learning

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Abstract

Collaborative learning composes defenses that were never designed together, so upstream processing can destroy what the downstream defense reads. The natural repair is to require the upstream stage to **preserve** it; for the broad class of *positive per-client rescaling* compositions, where the invariance question is exactly algebraic, that is not enough. The obstruction is structural: a criterion gated on the outcome it predicts conditions on a descendant of both the mechanism and the defense’s standalone strength, so **no screen of this form identifies the mechanism’s contribution, at any sample size**. Ignoring this does not cost power, it costs the sign: on the identical cell the outcome-gated ladder moves -0.272 where the instrument moves $+0.098$. Preservation is a hierarchy, and we break its last two links by intervening rather than comparing: Oracle intervention pinning every adversary’s coefficient closes the attenuation channel, and disturbing the statistic Krum reads then flips its decision in up to 80% of rounds while the *support* of the admitted adversarial mass never changes in any measured round of any arm, and suppression stays inside our pre-specified ± 0.15 equivalence margin. We prove sufficient conditions for the weighting→aggregator class and replicate the negative in a second invariance class and on a second dataset and architecture. The implication for decentralized training is methodological: evaluate composed mechanisms by intervening on adversarial contribution, not by inferring it from downstream statistics.

1 Introduction

Federated learning [McMahan et al., 2017] is vulnerable to data poisoning [Chen et al., 2017, Bagdasaryan et al., 2020], and an arms race has produced defenses [Fung et al., 2020, Pillutla et al., 2022, Yin et al., 2018, Blanchard et al., 2017] with accuracy/robustness trade-offs and none universally dominant (Appendix F). *Can we compose them for stronger robustness?* Composition does not even commute: FoolsGold→CoordMedian suppresses a pixel backdoor to ASR 0.076 while the reverse order reaches 0.732: a rank aggregator emits one vector rather than per-client updates, so it cannot be an *upstream* stage and the pair collapses to FoolsGold alone, bit-identically. Naive layering fails this way whenever upstream processing destroys what the downstream defense reads.

Why the question is sharper without a trusted coordinator. Such stacks compose independently developed mechanisms among heterogeneous, partially trusted, often pseudonymous participants, so preservation is the obvious discipline to demand of them. Concretely: a contributor clips its own update, a scheduler reweights contributors by reputation, and whoever merges runs a robust aggregator: three parties, three mechanisms, nobody owning the stack. The only check at each seam

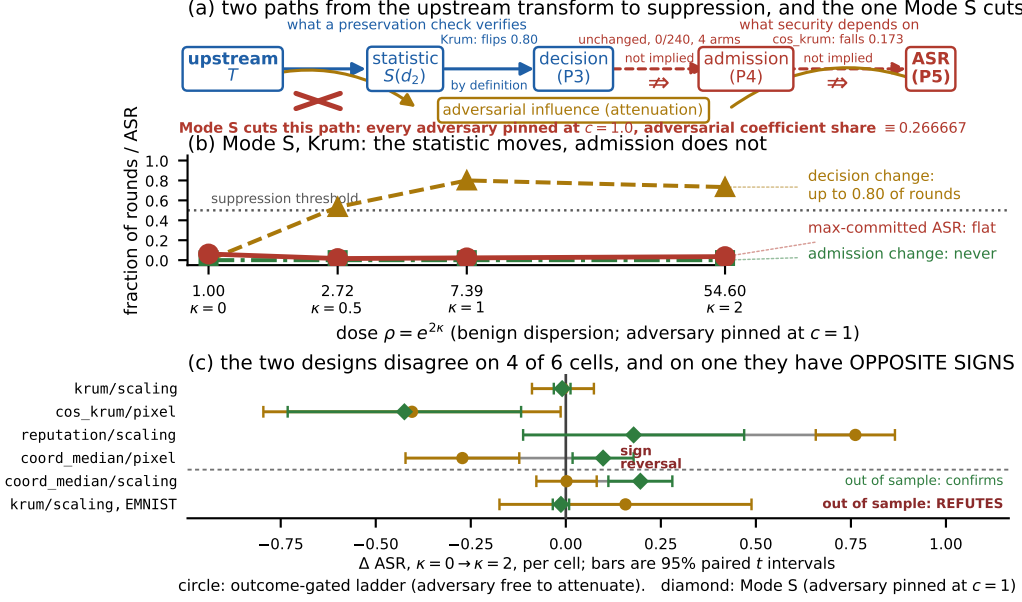


Figure 1: **The paper in one figure: the statistic moves, admission does not, suppression does not follow.** (a) Two paths run from an upstream rescaling T to suppression: the *chain* a preservation criterion reasons along, and a *confounded* path on which T merely attenuates whichever client carries the poison (Lem. 1) and reaches ASR **without passing through d_2 's statistic at all**. **Mode S cuts the second by construction**, pinning every adversary at $c=1$ (the one channel the dose closes, and not the only thing it moves). The admission count pools the four aggregators of §5; Krum admits no adversary at any rung, so its zero is a floor (App. C). (b) Mode S, Krum under model-scaling (§4): as ρ grows to 54.6 the *decision* flips in up to 80% of rounds while the *support* of the admitted adversarial mass does not, and max-committed ASR stays inside the pre-specified ± 0.15 equivalence margin ($|\Delta| = 0.026$). Error bars in this panel are Student- t 95% intervals on the levels ($n=5$). (c) **Six cells, one contrast each, estimated two ways** (circle: outcome-gated ladder; diamond: Mode S, adversary pinned at $c=1$). **The designs disagree on 4 of 6 cells, and on coord_median/pixel they have opposite signs:** $\Delta \text{ASR} = -0.272$ (paired 95% t interval $[-0.422, -0.122]$) against $+0.098$ ($[+0.018, +0.178]$), intervals excluding zero and not overlapping, so **closing the attenuation channel does not shrink the effect, it reverses its sign**. Above the dashed rule are the four cells a pre-registered mechanism was read off (*training data, not evidence for it*); below are its two out-of-sample cells, one of which **refutes it, so we withdraw it**.

is that the next stage still reads what it expects, so a preserved statistic is read as evidence the stack is robust. **That inference is what this paper refutes. The identification problem we isolate is defined independently of model scale**, so we study it in a controlled federated setting and claim no transfer beyond it.

Contributions. We propose no new defense, and **our central contribution is a negative identification result**: an outcome-gated comparison of defense pairs cannot determine whether preserving a downstream defense's mechanism is what keeps the composition secure. *Attack-suppression preservation* names the object throughout: suppression of the committed attack surviving composition, instantiated as ASR, never the statistic or the decision that follows; Fig. 1 is the paper in one picture. **The boundary.** Outcome-gated cross-defense screening is structurally unidentified: its eligibility gate is defined on the same outcome it predicts, so no cross-defense contrast *satisfying this outcome-gated design* identifies the mechanism's contribution and the admissible designs are within-defense (§3). **The demonstration.** Statistic preservation is accordingly not a valid causal proxy for attack-suppression preservation, established by intervention rather than correlation: on the identical cell the outcome-gated design estimates -0.272 where the intervention estimates $+0.098$, reversing the direction and not merely the precision (§4). **The consequence.** The screening rule follows from the boundary and is nothing more: evaluation prioritization, measurably broken under heterogeneity

(App. J.2). **Also.** The preservation hierarchy (Def. 1); mechanism preservation as invariance under positive per-client rescalings (Prop. 1), with margins on the weighting→aggregator class (Thm. 1), supporting machinery, safe at *both* extremes of the weight ratio, so dispersion alone mispredicts; counterexamples across four aggregators of three structural kinds; and the negative replicated in a second invariance class and on a second dataset and architecture, while refuting the cross-arm admission ordering we had predicted. Scope is inventoried once, in Tables 1 and 2: what each claim does *not* license, and the evidence tiers (App. J.1).

2 What Preservation Could Mean: A Hierarchy of Five Levels

What does it mean for a composition to preserve a defense? Every per-client defense we consider acts as a positive scalar rescaling $T(u_i) = c_i u_i$, which gives the invariance question a sharp *algebraic* answer (Prop. 1), but writing C2 as an invariance of the statistic S leaves a gap our own experiments exposed. Three different things can be preserved: what the downstream defense *reads* (P1–P2), what it *decides* and what *survives* it (P3–P4), and whether the attack is still *suppressed* (P5). They come apart, so we name the levels and ask of each whether it determines the one below:

Level	What is preserved	Determines the level below?
(P1)	the statistic’s value	yes, by definition
(P2)	the ordering it induces	yes, by definition
(P3)	the decision d_2 makes	no , by counterexample (§4)
(P4)	the adversarial contribution admitted	no , by counterexample (§5)
(P5)	suppression of the committed attack	—

Both breaks are measured, at fixed d_2 and attack: Krum’s decision flips in up to 80% of rounds with the adversarial mass it admits unchanged (§4), and `cos_krum`’s decision and admitted mass are bit-identical while ASR falls 0.173 (§5); Fig. 2 draws the chain and both breaks.

The algebra fixes which level a given composition sits at: cosine similarity is exactly invariant, so C2 holds for every $\cdot \rightarrow$ FoolsGold composition; coordinate ordering survives only under a bounded-ratio condition; consensus distance and residual magnitude are not invariant at all, so $\cdot \rightarrow$ Reputation compositions are C2 failures; norm clipping does not blind FoolsGold, though it does blind Reputation.

Definition 1 (Preservation levels). *For an upstream transform T and a downstream defense d_2 reading statistic S :*

- (P1) **Value invariance:** $S(T(U)) = S(U)$.
- (P2) **Ordering invariance:** $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces on clients. (this is C2).
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the pre-registered term throughout for how much adversarial input survives d_2 : a level, not one measurable quantity, instantiated by whatever each rule exposes: Krum’s and `cos_krum`’s selected-client indicator, `coord_median`’s selected-coordinate share, reputation’s adversarial weight share, which is why App. C’s influence column is aggregator-specific.
- (P5) **Attack-suppression preservation:** ASR stays below threshold.

We target (P4) $\nRightarrow$ (P3) in §4 and (P4) $\nRightarrow$ (P5) in §5. The conditions (C0)–(C3) the levels induce (**the rule being C1 $\wedge$ C2**) are in App. A and B.

What the theory delivers: supporting machinery, not the claim. For a per-client *weighting* into a *rank-based* or *geometric* aggregator, Thm. 1 gives sufficient conditions for d_2 ’s ordering to survive: it **certifies mechanism preservation, not security** and bounds no ASR. Its one prediction, admitted adversarial mass $\Lambda_a \leq 12.6\%$ against a realized $\leq 1.8\%$, goes vacuous as the separation margin approaches 1, which heterogeneity reaches, and does not follow from a (d_1, d_2) definition pair alone, as a screen would need (App. I).

96 3 The Testability Boundary

97 *Why can we not simply compare many defense pairs?* The natural test keeps pairs whose constituents
 98 already suppress the attack and asks whether the composition preserves it. That is exactly where
 99 identification is lost. **The boundary in words** (Prop. 4, formal in App. J.4): let a screen certify
 100 $(d_1 \rightarrow d_2)$ only if an *eligibility gate* holds: some constituent already suppresses the attack, measured
 101 by the same outcome metric and threshold the screen predicts for the composition, and a mechanism
 102 condition P holds, fixed by the two defenses’ definitions alone. Four cells follow, and that design
 103 never fills two of them:

	Mechanism condition	d_2 suppresses a alone	it does not
104	P holds (preserved)	observed: every certified pair	empty by the gate
	$\neg P$ (destroyed)	observed: predicted-FAIL pairs	empty by the gate

105 Reading down a column is the contrast that would identify P ’s contribution, and **the only column**
 106 **in which d_2 is not already doing the work is empty**, for every menu a screen of this form could
 107 range over: excluded by design, not undersampled: more data cannot fill a cell the design never
 108 permits. The gate cannot be dropped either, because without it C1 is vacuous. So wherever our
 109 conditions certify a pair, a low composition ASR is attributable to d_2 ’s standalone strength rather
 110 than to preservation. Equivalently, this is selection on a collider [Elwert and Winship, 2014, Pearl,
 111 2009]: the gate is a descendant of both P and d_2 ’s standalone effectiveness, so conditioning on it
 112 makes them co-vary. The hazard is standard; the gate reading as hygiene rather than as the step that
 113 removes identification is not, and we made that mistake ourselves (§5).

114 **The admissible design, derived: the practical contribution.** Read forwards the boundary is
 115 *constructive*, each clause forced by a term in Prop. 4: (1) hold d_2 and the attack fixed, so the gate’s
 116 input is the same quantity on both sides of every contrast; (2) intervene on the upstream transform
 117 rather than selecting a different d_2 , whose variation is what empties the cell; (3) pin the adversarial
 118 coefficient channel, since an upstream that attenuates the adversary moves the outcome by a route
 119 unrelated to d_2 ’s statistic, the confound that inverted our own sign; (4) measure every channel:
 120 aggregate displacement, decision, admission, influence. §4 is clause (3) as an instrument; the six-
 121 step form is in App. J.3. A second boundary bounds what such a screen can *find*: **no emergent**
 122 **composition can satisfy an inherited-suppression gate**, our own lowest-ASR composition being
 123 the witness: FG→RFA reaches 0.093 where FoolsGold alone reaches 0.732, so at $|\mathcal{E}|/|\mathcal{S}|=1/5$ the
 124 bound caps recall at 80%, above the 40% we measure (App. J.2).

125 4 Statistic Preservation Does Not Identify Attack-Suppression Preservation

126 *Does disturbing what a defense reads actually cost suppression?* The claim: it need not, and **the**
 127 **design that says otherwise gets the sign backwards**. Our pre-registered pooled hypothesis (H1),
 128 that ASR rises with disturbance of the statistic d_2 reads, is rejected, the effect *negative*: that ladder
 129 moved adversarial influence and the statistic together (§5). We adjudicate C2’s implication that
 130 losing (P3) means losing (P5) by oracle intervention on two arms in two invariance classes, not by a
 131 general law.

132 **The instrument, as a counterfactual.** The clean question asks what happens when the down-
 133 stream statistic changes but the adversary cannot become weaker. **Mode S** enforces exactly that: it
 134 pins every adversary’s coefficient at $c=1.0$ and spreads only the benign clients over $\rho = e^{2\kappa}$, hold-
 135 ing the adversarial coefficient share constant at 0.266667 to within float32 read-back noise, closing
 136 Figure 1(a)’s attenuation path, so a rise in ASR here cannot be attenuation. Mode S and Mode A (§5)
 137 are causal instruments built from adversary identity: they measure mechanism-level effects rather
 138 than emulate deployable defenses (Table 1, App. J.5).

139 **Two readings of C2, frozen against each other.** *The statistic reading*: what matters is disturbance
 140 of d_2 ’s statistic, read off its *decisions*. *The admission reading*: what matters is the *adversarial input*
 141 d_2 *admits*. Mode S disturbs Krum’s selection heavily while leaving the mass it admits untouched,
 142 so that arm adjudicates (Table 4, the pre-registration’s H-statistic and H-admission): as ρ climbs
 143 to 54.6 we find the decision changing in up to 0.80 of rounds while the adversarial mass admitted

changes in none of them, and ASR showing no evidence of the predicted increase, remaining inside the pre-specified equivalence margin ($p_{\uparrow}=0.70$, $|\Delta|=0.026 < 0.15$). **The statistic reading is refuted:** statistic and decision disturbance need not change suppression once the adversarial coefficient share is held fixed. *On this arm the zero is a floor:* Krum admits no adversary at any rung, so the dissociation is read across all four aggregators, each under its own committed attack:

Aggregator	Kind	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector	0.733	0.000	0.000	-0.026
Reputation	weighted averager	1.000	0.000	0.020	+0.178
Cosine-Krum	cosine-invariant selector	0.000	0.000	0.000	-0.425
Coord. median	coordinate order statistic	0.482	0.000	0.033	+0.098

Δ adm. is the *support* of the admitted adversarial mass: it never moves in any of the 60 measured rounds of any arm, including the 48 per arm where reputation and coord.median carry adversarial mass at baseline. $\Delta\Lambda_a$ is the separate question of how much weight the adversary’s direction carries. cos_krum’s -0.425 is the full-ladder endpoint, below our accuracy floor; the witness we rest on is its first, gate-passing rung (0.173, §5); App. C adds EMNIST-byclass and the score-only control.

What the instrument does not hold fixed, and the pre-registered control that closes it. Mode S does not identify the effect of statistic disturbance in isolation: it identifies the effect of the upstream intervention while holding the adversarial coefficient share fixed, which is what lets statistic and decision disturbance be *dissociated* from suppression rather than shown to be inert. Mode S pins the *adversarial* coefficients and nothing else: the dose also moves Krum’s pairwise distances, *which* benign client it selects, and the **magnitude and geometry of the update actually aggregated**. That third channel is measured: App. C’s Δ agg. column, the aggregate the defense *emits*, reads **0.892** here (App. D). **Score-only Krum** closes it: Krum scores the transformed stack and so makes the same selection, but the *untransformed* selected update is aggregated: the decision channel preserved, the magnitude and geometry change removed. Rules were frozen at 35788d9 before any score-only ASR existed, $\Delta > +0.15$ meaning the published result was partly a magnitude artifact. We find $\Delta = -0.023$ ($0.062 \rightarrow 0.039$) against -0.026 uncontrolled on the identical cell, at the same Δ dec. of 0.733, the refuting threshold 0.212 nowhere approached and every rung clearing the accuracy floor, so the central claim is instrumented against two independent confounds, attenuation and magnitude (App. E).

A pre-registered replication that refutes the positive half, and the sign reversal. coord.median under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the smallest decision change (0.482) and the largest admission-level change ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise, and thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$). We find $\Delta = +0.098$, which **refutes the admission ordering**: the arm with the largest admission-level change rose *less* than one with a smaller change. The decision reading does not predict it either ($R^2=0.239$ pooled). What replicates is the negative on a coordinate-wise order statistic: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698. Identity ASR here is 0.443, so equivalence is to the *identity rung*; the rise is positive in 5/5 seeds (post-hoc): small and consistent, not zero.

The confounded design gets the causal sign wrong. On this identical cell (matched displacement, decision change and clean accuracy, Table 3), closing one channel reverses the sign:

Design	The adversarial coefficient	Δ ASR
Outcome-gated ladder (§5)	free to attenuate	-0.272 ($p_{\downarrow}=0.0015$)
Mode S intervention (§4)	pinned at $c=1$	+0.098

The confounded design does not lose power, it gets the sign wrong. Closing the attenuation channel inverts the effect rather than sharpening it: a criterion calibrated against the confounded contrast would have been calibrated against the confound, so §3’s boundary is not a technicality about power. **Downstream-statistic preservation is not, in general, sufficient to establish attack-suppression preservation** (Prop. 2): no predicate of the statistic alone determines the admitted adversarial mass, and none of that mass alone determines suppression: the title claim, by counterexample *within this class* (Table 1).

5 What the Boundary Licenses, and What We Found

We validate on CIFAR-10 with a 4-layer CNN, $N=10$ clients, $f=0.2$ adversarial, two committed attacks (model-scaling, pixel backdoor), at $n=5$ seeds per cell; *ASR* means *max-committed ASR*; the rest is fixed in App. X. The core evidence is a seed-matched intervention on one cell, not a mean comparison across cells.

The confounded ladder that motivated the instrument, and the sign it got wrong. Our earlier suites tried to validate C2 by *choosing a different downstream defense*, the move Prop. 4 rules out. Varying only the *dose* at fixed d_2 and attack ($\kappa=0$ bit-identical to d_2 standalone) refutes that suite’s pooled prediction that disturbance predicts each arm’s rise: $\rho_s=+0.351$, one-sided $p=0.091$ against a frozen $p < 0.05$, and in three of four arms *ASR falls* as the dose rises (80 runs, one arm confirming decisively; Table 5). **The fault is ours and is measured:** at fixed mean(c) a large spread mostly *dilutes* whichever client carries the poison, so this is attenuation, not statistic disturbance (App. M, R).

The dissociation holds across four aggregators and replicates on a second dataset. Applying the instrument to that ladder, most of the one confirmation was attenuation: with the adversary’s share pinned, reputation’s rise at $\rho=54.6$ shrinks from $+0.762$ to $+0.178$, and what remains tracks a small *admission* change, not the large decision change (App. J.5). The class-(a) control refutes us the other way: *cos_krum*’s selection is bit-identical at every rung yet *ASR* is 0.584 at the identity (it never suppresses) and *falls* 0.173 at the first accuracy-gate-passing rung (one-sided $p=0.044$, 5/8 seeds); the full-ladder fall sits below our accuracy floor, so **the gate-passing rung is the witness**. **Mode A** (payload share at fixed dispersion) separates Thm. 1’s two regimes: confirmed for *coord_median*, refuted for reputation. A *pre-registered* replication on EMNIST-byclass/simple_cnn moves the decision on 0.643–0.714 of rounds with the adversary never admitted and suppression inside the frozen ± 0.15 margin ($\Delta = -0.017$ beside CIFAR-10’s -0.026): **replicated** on a second dataset and architecture, never “generalizes” (App. C, K).

And what a screen then buys. Only the 13/20 prospective sub-tier risks falsification, limited by our menu: FLTrust suppresses both attacks alone (0.048), so its pairs are out of scope or near-identities, and two further attempts to validate the *screen* were defeated by the same confound. **A third closed a census.** The region Prop. 1(a) settles *exactly* is 3×3 ; its 2 unspent cells and 2 measured-C2-fail controls were labelled at 96f8856, and **all four landed as registered**. It holds **one** low-*ASR* pair, $\text{rep} \rightarrow \text{cos_krum}$ at 0.263, which neither constituent suppresses alone, **and the criterion selects it:** C2 alone admits 9 (11%, 90 runs), $C1 \wedge C2$ admits 1 (100%/100%, 10). **Where invariance is exact the trade reverses**, on one accuracy-gated cell at $n=3$ whose C1 input is itself seed-count-contingent (33% ungated; App. K.2). **Scale:** the PASS predictions hold at $N=100/\text{ResNet18}$, far from converged accuracy. Under *one* adaptive stress test $\text{FG} \rightarrow \text{RFA}$ rises $3.7\times$: **inherited robustness is not adaptive robustness** (App. K, O, S).

6 Conclusion

The finding. Downstream-statistic preservation is not sufficient to identify preserved attack suppression. An intervention changes a defense’s statistic and decision substantially without changing the adversarial mass it admits or the suppression that follows; conversely a bit-identical statistic and decision coexist with a change in suppression, each direction with a measured witness.

Why. An outcome-gated cross-defense comparison conditions on the same outcome it seeks to explain, so it cannot separate mechanism preservation from standalone effectiveness: a structural identification boundary (Prop. 4) no amount of additional data within that design removes. Within-defense intervention fixes d_2 and the attack instead: on the identical cell **the confounded design does not lose power, it gets the sign wrong**.

The implication. Evaluate composed defenses by measuring the adversarial contribution that survives the downstream mechanism, not by inferring security from the statistic it consumes. That object is (P4), which Mode S reads only by knowing adversary identity, so we identify the channel rather than screen for it (App. J.3). The open question: **can (P4) be estimated without knowing who is malicious?**

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292 Table 1 sits ahead of the appendices because it governs all of them: it is the one place each scope
293 condition is stated, and no later section restates them at each use.

Table 1: **Scope of claims.** The one place each scope condition is stated. **Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them.

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution; admissible designs are within-defense (Prop. 4, Prop. 5).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that a composition <i>is</i> secure, nor that mechanism preservation implies suppression.
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators (Thm. 1, Lem. 1, Cor. 1).	Proved on that class only: a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first has no such T and is covered by nothing here (C0, applied retrospectively to arms frozen before it). Bounds no ASR , and Regime A’s condition is not operationally checkable (S is not a stable per-attack constant).
Empirical	§4’s interventions and §5’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha=0.5$, star topology , three attack families; one arm on EMNIST-byclass/simple_cnn.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos_krum</code> , $n=3$ on EMNIST-byclass), fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered equivalence margin. Dataset transfer is tested on the flagship arm alone: “replicated on a second dataset and architecture,” never “generalizes”. For the screen, dataset transfer on EMNIST-byclass is out of scope rather than merely untested: no single defense measured there suppresses the committed pixel backdoor alone (minimum 0.609), so C1’s per-attack precondition fails over the menu measured (§K.3). “Refuted” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved.
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c .	They identify a causal channel and are not screens ; computing (P4) without adversary identity is our open problem, and admission is a <i>direction to measure</i> , not a validated predictor. C1 and all built on it assume a committed adversary: what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

294 A The composition model and the invariance classification

295 **The two witnesses under Definition 1, and which rung each rests on. (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) hold**
296 **by definition: logical implications, not empirical claims;** their content is that they stop at (P3), and
297 §2’s table states both breaks below it. *Holding* decision *and* admission invariance does not deliver
298 suppression: `cos_krum` falls 0.173 by the first rung alone through the magnitude channel, at $\rho=2.72$,
299 where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (Appendix N).
300 Across the whole ladder it falls 0.425, but that endpoint sits below our accuracy floor, so **the gate-**
301 **passing rung is the witness we rest on** (§5). *Losing* decision invariance does not cost suppression:
302 Krum’s selection flips in up to 80% of rounds under an intervention pinning the adversarial mass, and
303 its ASR stays inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) (§4). So a criterion
304 stated at (P1)–(P2) constrains security in *neither* direction, **(P4), where the security content sits, is**
305 **not sufficient for (P5) either, and the hierarchy answers “which level does a security claim have**
306 **to be stated at?”, not “which level predicts ASR?”**

307 **Definition 2** (Defense composition). *Given defenses d_1 and d_2 , composition ($d_1 \rightarrow d_2$) operates*
308 *as: d_1 produces per-client transformed updates $U' = \{T_{d_1}(u_i)\}_{i=1}^K$; d_2 aggregates U' using its*
309 *standard algorithm. For FoolsGold $\rightarrow$ CoordMedian, FG outputs $\{w_i u_i\}$ and CoordMedian takes*
310 *the unweighted coordinate-wise median of the K transformed values, not a “weighted median”.*
311 *Clients with $w_i = 0$ are **not removed** but reach d_2 as zero vectors, so at least one point of U' is*
312 *always the origin, which is what Lemma 1 is about.*

313 Each defense suppresses malicious updates through a *discriminative property* (a structural feature
314 separating adversarial from benign contributions) and the property is defense-class-specific: *simi-*
315 *larity spread* (FoolsGold), *coordinate ordering* (TrimmedMean, CoordMedian), *residual magnitude*
316 (RFA), *consensus distance* (Reputation, in ℓ_2 from the coordinate-wise median) and *norm magni-*
317 *tude* (NormClip). Mechanism preservation is then governed by whether d_1 ’s transform leaves d_2 ’s
318 property intact, and because every per-client defense we consider acts as a positive scalar rescaling,
319 that question has a sharp answer:

Proposition 1 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, strictly: the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \varepsilon)$), each positive by construction. **FoolsGold is deliberately not in that list:** its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of 5 clients here), so FoolsGold-as- d_1 falls outside the hypothesis and is governed by Lemma 1 rather than by (a)–(c). Then:*

- (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. Hence FoolsGold’s weight vector is unchanged, $w(T(U)) = w(U)$, and **C2 holds for every $\cdot \rightarrow$ FoolsGold composition.**
- (b) **Coordinate ordering is not invariant in general;** it survives under the bounded-ratio condition of Theorem 1(1).
- (c) **Consensus distance and residual magnitude are not invariant:** heterogeneous c_i can invert their ordering, so $\cdot \rightarrow$ Reputation compositions are C2 failures and $\cdot \rightarrow$ RFA requires the separation condition of Theorem 1(2).

So mechanism preservation is a question about the *invariance class* of the downstream signal, and the classification corrects a natural but wrong intuition: norm clipping does not blind FoolsGold, though it does blind Reputation, whose signal is consensus distance rather than similarity.

The framework in general form. The structure does not depend on rescaling. Write d_2 ’s discriminative property as a statistic $S(U)$ and let T be d_1 ’s per-client transform; then $(d_1 \rightarrow d_2)$ is **mechanism-preserving** when $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering d_2 acts on. C2 is exactly that requirement, C3 asks in addition that φ not compress the gap below d_2 ’s operating threshold, and Proposition 1 is the case $T(u_i) = c_i u_i$ (Appendix I), so the conditions are algebraic rather than experimental. It also states the real limit: **a d_1 whose transform is not per-client (mixing updates, or carrying cross-round state) has no T of this form, and nothing here applies.**

The full causal chain. Definition 1 is a prefix of a longer one: statistic $\rightarrow$ decision $\rightarrow$ admission $\rightarrow$ aggregate adversarial influence $\rightarrow$ per-round update $\rightarrow$ persistence $\rightarrow$ ASR. **We measure the four emphasized links, not update geometry or persistence dynamics:** an acknowledged gap, not an assumed-benign one. A selection flipping from one benign client to another disturbs the statistic maximally and the mechanism not at all, which is why (P3) and (P4) come apart. **“Mechanism” always means the downstream defense’s discriminative operation, never its security behaviour,** that implication being what this paper refutes.

B The signal-preservation conditions in full

Definition 3 (Signal-Preservation Conditions). *Composition $(d_1 \rightarrow d_2)$ is predicted to suppress attack a when: (C0) d_2 ’s mechanism is load-bearing, $\text{ASR}(d_1, a) \geq 0.5$ for some committed a : a condition on the criterion’s domain that **never certifies a pair**, and one we formulated only after §5 made it bind, so it is applied **retrospectively** and is **not part of what the 42 pairs validate**; (C1) some constituent suppresses every committed attack alone, $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$ for all a , defined through the same threshold that constitutes the outcome, hence an **empirical prerequisite and not a mechanistic property**; (C2) d_1 ’s transform does not destroy d_2 ’s discriminative property, decided from definitions alone by Proposition 1; (C3) it does not compress the quantitative margin below d_2 ’s operating threshold. **The prioritization rule is $C1 \wedge C2$, C3 demoted to a margin diagnostic (Appendix U); the pre-registered labels of §5 were frozen with C3 in the gate and are not relabelled. Theorem 1 bounds no ASR: that mechanism preservation plus C1 predicts low ASR is a separate empirical hypothesis, tested in §5, and neither stage alone establishes low ASR, nor does their conjunction (stage structure, Appendix B).***

Definition 3 in the body states the four conditions and every disclosure attached to them; what follows is the same criterion with its *stage structure*, which is worth separating because conflating the stages is the main source of confusion about what the framework claims.

Composition $(d_1 \rightarrow d_2)$ suppresses attack a when the conditions below hold. The criterion is a **two-stage screen**, and we label the stages because conflating them is the main source of confusion about what the framework claims: stage 1 is empirical screening of the *constituents*, stage 2 is mechanistic screening of the *composition*.

372 **Stage 0: applicability (C0).** Whether the framework has anything to say about this pair at all. This
373 is a condition on the criterion’s *domain*, and Lemma 2 is stated over it together with C1.

374 **(C0) The downstream mechanism is load-bearing.** $\text{ASR}(d_1, a) \geq 0.5$ for at least one committed
375 attack a . C2 and C3 concern preservation of d_2 ’s mechanism, so they carry information only when
376 that mechanism does work the composition needs: if d_1 alone suppresses everything, the composi-
377 tion inherits that suppression and a correct prediction is correct for a reason the framework did not
378 supply. The asymmetry is deliberate: a strong d_2 is the intended case. **Failing C0 removes a pair**
379 **from the criterion’s domain; it never certifies one.** *Provenance:* C0 was vacuous on the menu
380 the framework was developed on (strongest single defense 0.519) and was formulated only after the
381 prospective suite of §5 made it bind, so it is applied **retrospectively** and is **not part of what the 42**
382 **pairs validate** (Appendix Q).

383 **Stage 1: empirical constituent screening (C1).** Read off standalone evaluations. This screens; it
384 does not explain, and it is not a mechanistic claim.

385 **(C1) Individual effectiveness.** For *every* committed attack a in the evaluated set,
386 $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$. The quantifier matters: our headline metric is the *maximum*
387 over attacks, so requiring a suppressing constituent for only one attack would license compositions
388 that fail on the other. C1 is defined through the same ASR threshold that constitutes the outcome,
389 so it is an **empirical prerequisite, not a mechanistic property**: the one condition that cannot be
390 checked without running something.

391 **Stage 2: mechanistic composition screening (C2, with C3 as a diagnostic).** Determined by d_1 ’s
392 and d_2 ’s definitions alone, no experiment required, and what Theorem 1 proves. **The evaluation-**
393 **prioritization criterion is $C1 \wedge C2$,** with C3 retained as a **margin diagnostic** rather than a gate for
394 the reasons scored in Appendix U. *The pre-registered labels of §5 were frozen with C3 in the gate*
395 *and are not relabelled*, and the demotion changes no reported number.

396 **(C2) Discriminative-property preservation.** d_1 ’s transform does not destroy d_2 ’s discriminative
397 property, as determined by Proposition 1: automatic for cosine-similarity downstream defenses, condi-
398 tional for coordinate ordering, and violated for consensus distance under heterogeneous rescaling.

399 **(C3) Margin preservation.** d_1 does not compress the quantitative margin below d_2 ’s operating
400 threshold. For rank aggregators: extremeness gap survives reweighting. For geometric aggrega-
401 tors: separation condition holds on transformed points.

402 **What is proved and what is hypothesised.** Stage 1 is empirical and outcome-defined, stage 2
403 mechanistic, and **neither alone establishes low ASR, nor does their conjunction**: Theorem 1 es-
404 tablishes that $C2 \wedge C3$ preserve d_2 ’s discriminative mechanism on the reweighted inputs and **bounds**
405 **no ASR**. That mechanism preservation plus C1 predicts *low ASR* is a separate empirical hypothesis,
406 tested in §5.

407 C The four channels, all four aggregators

408 Endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S, for every
409 downstream defense in the paper. **The first column, $\Delta \text{agg.}$, is the relative displace-**
410 **ment of the aggregate the defense emits,** $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$: what enters
411 training, and **not** the value of a scoring statistic. The measured channels are therefore ag-
412 gregate displacement, decision, admission and influence. **Recomputed, not transcribed:**
413 every cell is emitted by experiments/build_channel_table.py from admission_
414 measurement.json, feminist_admission.json, targeted_dose/summary.json, dose_
415 replication/summary.json and dose_femnist/summary.json under results/, which
416 the script only reads; it also asserts that the identity rung displaces nothing (0.00e+00 for all
417 six rows). **The sixth row is the score-only control** (Appendix E): the first row’s arm with the
418 magnitude channel closed, so its aggregate column is the re-selection component alone (from
419 displacement_decomposition.json) and its decision, admission and influence columns are
420 the first row’s own numbers, score-only leaving the selection untouched; its ASR comes from
421 score_only/summary.json, which the emitter refuses to merge into the ASR sources it shares
422 cell keys with, so it cannot shadow the first row’s published number. The influence column is
423 **aggregator-specific by necessity** (a selector has no weights) and its instantiation is: adversarial

424 share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected
 425 coordinates for the order statistic, and the indicator that an adversarial update is selected at all for
 426 the two selectors. Earlier drafts labelled this arm “FEMNIST” in the tables while the prose said
 427 EMNIST-byclass: the loader reads torchvision EMNIST(split="byclass") under a femnist key
 428 and partitions it by the label-Dirichlet split used throughout, *not* LEAF’s by-writer split, so the
 429 tables now read EMNIST-byclass while the frozen artifact and pre-registration filenames keep the
 430 femnist key. The EMNIST-byclass row is the second-dataset replication of §5; its channels were
 431 measured prospectively, before its ASR existed.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	−0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
432 Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	−0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (EMNIST-byclass)	selector (pairwise distance)	0.842	0.643	0.000	0.000	−0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	−0.023

433 **What this table does and does not show.** It shows that the dissociation between the decision
 434 channel and the admission channel is not a property of one aggregator: it holds for a selector, for a
 435 cosine-invariant selector, for a coordinate-wise order statistic and for a weighted averager, on two
 436 datasets. It does **not** order the arms ($\Delta\Lambda_a$ and Δ ASR disagree in rank, which is the refutation of §4
 437 stated in one row-pair) and it is *not* an out-of-sample test: every row is a Mode-S ladder reported
 438 elsewhere in this paper.

439 D Decomposing the emitted displacement: re-selection against rescaling

440 The Δ agg. column above mixes two things the causal claim needs apart. For a selector, $\text{agg}(U) =$
 441 u_{s_0} and $\text{agg}(T(U)) = c_{s_1} u_{s_1}$, with s_0 the client selected on the raw stack and s_1 the client selected
 442 on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

443 both components reported relative to $\|u_{s_0}\|$. They are **not orthogonal** (the measured angle runs
 444 from $\cos = -0.99$ to -0.43 , mean ≈ -0.65), so their norms need not sum to the total, and the
 445 table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead
 446 is the exact vector identity: worst residual $2.66\text{e-}07$ over all 240 (arm, family, rung, seed, round)
 447 cells, float32 noise. **This decomposes the published number, not a second similar one:** every
 448 recomputed total agrees with the corresponding Δ agg. value in `admission_measurement.json`
 449 to $0.00\text{e}+00$, on the same seeds, rounds and configuration. Selectors only: a weighted averager
 450 and a coordinate-wise order statistic emit no single selected client, so the split is undefined for
 451 them. `experiments/measure_displacement_decomposition.py` trains no ASR, writes a new
 452 file, and leaves every frozen artifact byte-identical.

Design	Selector	κ	total	re-selection	rescaling	cos	sel. chg.
Confounded ladder	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.637	0.685	0.333	−0.65	0.600
		1	0.731	0.851	0.685	−0.66	0.733
		2	0.880	1.675	1.674	−0.72	0.867
Instrument (Mode S)	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.663	0.646	0.308	−0.64	0.533
		1	0.847	1.075	0.666	−0.69	0.800
		2	0.892	0.986	0.973	−0.66	0.733
Confounded ladder	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.270	0.000	0.270	—	0.000
		1	0.515	0.000	0.515	—	0.000
		2	0.834	0.000	0.834	—	0.000
Instrument (Mode S)	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.395	0.000	0.395	—	0.000
		1	0.754	0.000	0.754	—	0.000
		2	1.297	0.000	1.297	—	0.000

What it says, and what it does not. Mode S into Krum at $\kappa=2$: the published 0.892 is 0.986 of re-selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and *how large* it arrives, in comparable measure. Of the 11/15 rounds whose selection changes there, **none** lands on an adversary: the quantitative form of the claim that Krum’s decision changes are benign-to-benign, and the reason the support of the admitted adversarial mass changes in **none of the 60 measured rounds** while the decision channel moves by 0.733, on this arm no adversary is admitted at *any* rung, so that count is a floor, and the arms carrying adversarial mass at baseline are reputation and `coord_median` (48 of 60 rounds each, App. C). `cos_krum` is the informative contrast: its statistic is exactly invariant to positive rescaling, so 0/15 selections change at every rung of *both* designs and its entire displacement (1.297 under the instrument, 0.834 under the ladder) is rescaling. Its fall therefore cannot travel through the decision channel at all, which corroborates §5’s attribution of that fall to the magnitude channel from a quantity measured independently of any ASR. **This is a decomposition of a displacement, not of an ASR effect:** it says what the dose moves in the update entering training and makes no claim about suppression, and it is an analysis of runs already reported rather than a new experiment.

E The score-only control in full

§4 reports the primary outcome: $\Delta = -0.023$ (mean ASR $0.062 \rightarrow 0.039$, seeds 42–46) against -0.026 for the uncontrolled arm on the identical cell, inside the frozen ± 0.15 equivalence margin, with the accuracy floor cleared at every rung (0.565/0.629/0.624/0.630). The verdict wording is the pre-registration’s own for that branch: *the negative survives with the magnitude channel closed*.

What the arm is. Krum scores on the transformed stack (statistic and decision disturbed exactly as in the uncontrolled arm, since the scoring input is the same) and aggregates the *untransformed* selected update, so what enters training is bit-for-bit what a client produced. It is the reviewer-suggested control in its own terms: the selected update’s magnitude and direction are held fixed while the selection statistic is perturbed independently. Implementation is a defaulted `score_only=False` kwarg, so no existing call site changes and every frozen computation stays bit-identical; the suite’s own harness check verifies that on the published arms. The $\kappa=0$ rung is *imported* rather than re-run, because at the identity the transform returns the stack unwrapped and score-only Krum *is* Krum alone: an exact claim, verified bit-identically at one seed before the arm started.

Three disclosures, from the freeze rather than from hindsight. (i) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted flatness with attenuation already closed by construction) and the observed trend is a *fall* ($J=58.5$, $z=-1.110$, $p_{\uparrow}=0.867$, permutation $p_{\uparrow}=0.874$). We report it and do **not** score it as support for flatness; the pre-registration says in advance that at $n=5$ a null here is not evidence of flatness. (ii) The branches had unequal headroom. With the identity rung at 0.062, refuting us required mean $\text{ASR}(\kappa=2) > 0.212$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch

($\Delta < -0.15$) was *arithmetically unreachable*: ASR cannot fall below 0, so the largest possible fall on this cell is 0.062. (iii) **Selectors only.** A weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” is undefined for them and the control does not cover them: the `coord_median` arm carrying the +0.098 rise is therefore **not** magnitude-controlled, and neither is reputation’s +0.178. Score-only Krum is also an instrument and not a defense: it aggregates an update its own scoring stage never saw. And closing the magnitude channel still does not hold the model trajectory fixed, because which client is selected continues to change, by design, that being the channel left open.

The `cos_krum` corollary, and exactly what it settles. Section 5 of the same pre-registration asserted, before the run, that score-only `cos_krum` must be *bit-identical* across rungs: its statistic is exactly invariant under positive rescaling, so the selection cannot move, and with the emitted update untransformed the whole trajectory must repeat. It does (accuracy 0.528400 and ASR 0.003111 at all four rungs, $|\Delta| = 0.00\text{e}+00$ throughout) and the $\kappa=0$ value reproduces the published cell at that seed exactly, so the two pipelines agree at 0.000000 rather than approximately. **What the check does not do is confirm the arm’s published fall.** It runs one seed, as frozen, and that seed is unrepresentative of the arm: under the uncontrolled dose its own ladder *rises* +0.224 where the eight-seed mean *falls* −0.425. Closing the magnitude channel removes all 0.224 of that seed’s movement, which corroborates the *channel*, for `cos_krum` the emitted displacement is entirely rescaling (Appendix D), so no other route is open, but the attribution of the arm’s fall rests on that decomposition rather than on this check, and settling it empirically would need the arm’s other seven seeds. Expanding a pre-registered seed count after seeing the data is what pre-registration exists to prevent, so we report the invariance as confirmed and the arm-level attribution as **corroborated mechanically, not established at the arm mean.**

F Related work

The contrast in one sentence. Prior work asks whether a defense withstands an attack, and the composition literature treats composition quality as an *outcome to predict*. We ask a different question: the **identifiability of mechanistic explanations** for that outcome, under the evaluation designs commonly used to compare defenses. Concretely, whether an evaluation criterion can causally attribute a composition’s security to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum [Blanchard et al., 2017], Trimmed Mean [Yin et al., 2018], RFA [Pillutla et al., 2022], FoolsGold [Fung et al., 2020], FLTrust [Cao et al., 2021] and FLAME [Nguyen et al., 2022] are each evaluated as a single stage; we propose no new defense and reason about *ordered pairs* of existing ones. Ensemble adversarial training [Tramèr et al., 2018] *assumes* upstream processing preserves downstream signals; we make that assumption explicit, checkable, and (as it turns out) **the wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense [Shejwalkar and Houmansadr, 2021, Bhagoji et al., 2019], as do distributed and durable backdoors [Xie et al., 2020, Wang et al., 2020, Zhang et al., 2022]. C1 quantifies over a *committed* attack, so this literature bounds what any inherited-suppression criterion (ours included) can promise; our single whitebox stress test (§6) measures that gap rather than defending the criterion.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive [Croce et al., 2021]. Our point is methodological in the same way but about **identification** rather than attack strength: composition studies gate candidates on standalone effectiveness [Kairouz et al., 2021] and then read the composition’s outcome as evidence about the mechanism, which Proposition 4 shows cannot work at *any* sample size. What is new is the level hierarchy at which the preservation intuition fails and the design class that can test it (Appendix J.4).

The attack side of the same literature engineers *persistence* rather than magnitude: model-scaling [Bagdasaryan et al., 2020] and Neurotoxin [Zhang et al., 2022] both aim at triggers that survive rounds of clean aggregation, which is the quantity we measure as γ (§1) and the reason temporal defense switching does not help in our regime (Appendix O). Nothing in this paper proposes a new aggregator, a new attack, or a new defense; the object of study is the *composition* of defenses

that already exist, and the contribution is an identification argument about how such compositions can and cannot be evaluated.

G Notation

symbol	meaning
c_i	upstream coefficient applied to client i 's update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack's coordinate extremeness (Theorem 1(1))
m	RFA separation margin, $\Delta'_{\text{sep}}/2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

H Deployment guidance in full

Deployment guidance. (0) Check C0: if one candidate defense already suppresses every attack you have, deploy it alone: composition has nothing to add. (1) Check C1: does at least one defense suppress known attacks? (2) Check C2 by invariance class (Proposition 1): a cosine-similarity downstream defense has its signal preserved behind any positive rescaling, a consensus-distance one does not. (3) For weighting→aggregator compositions Theorem 1 gives verifiable sufficient conditions for C2∧C3. Gate on C2; read C3 as a *quantitative diagnostic*: a 5–10 round pilot returns ρ and the separation margin m , and it is $S > \rho$ (rank aggregators) or $m \gg 1$ (RFA) that matters, not whether d_1 happens to be heterogeneous, the binary reading that produced both of our false negatives. (4) Best candidate on our evidence, for further evaluation rather than deployment: **FG→CM** (accuracy 0.68, max ASR 0.107). FG→RFA is marginally lower on the point estimate (0.093 at $n=30$) but statistically indistinguishable, and carries the *highest* adaptive ASR (0.167) and an 18-point accuracy cost, so it is not the more secure choice on our evidence. (5) In our persistent-backdoor setting fixed-order composition substantially outperforms randomized ordering; we have not tested regimes where randomization might pay.

I Proofs

What the two arms jointly establish. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 2 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass (indeed the same statistic and the same decision) while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Both parts are proved by exhibited witnesses (Appendix I): Mode S into Krum under model-scaling for (a), cos_krum for (b). **The corollary is the paper's title claim:** no metric defined *solely* on preservation of the downstream statistic can identify attack-suppression equivalence, (a) exhibiting statistic disturbance without suppression change and (b) the converse. **Scope:** a non-existence claim *within the positive-rescaling class on these defenses and attacks*, by exhibited counterexample: **not** that no metric of any kind can order security.

Proposition 3 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic's dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

584 This makes §2’s warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**. It does
 585 *not* follow that Δ_c is a predictor; its magnitude fails to order suppression either (§5). The instrument
 586 is elaborate because §3 leaves nothing simpler. §5 answers what remains: *what happened when we*
 587 *tried it the obvious way first?*

588 I.1 Formal Guarantee: Weighting $\rightarrow$ Aggregator Class

589 For the practically important class containing our certified compositions (a per-client *weighting*
 590 defense d_1 followed by a *rank-based* or *geometric* aggregator d_2) C2 and C3 can be formally estab-
 591 lished. **Read what follows as a *boundary* theorem and deliberately not a *security* theorem**: it
 592 says exactly how much an invariance argument about composition can prove, which is preservation
 593 of d_2 ’s discriminative mechanism at levels (P1)–(P3). It bounds no ASR, and the counterexamples
 594 in the box above are counterexamples to reading it as though it did.

595 **Theorem 1** (Bounded reweighting preserves the downstream discriminative mechanism). *Let d_1*
 596 *apply per-client scalar weights $w_i \geq 0$ with ratio $\rho := \max_i w_i / \min_{i:w_i>0} w_i$ (over strictly positive*
 597 *weights), operating on n_b benign and n_a adversarial updates in a round of size K . Assume $w_a > 0$*
 598 *for the adversarial clients; the $w_a = 0$ case is Lemma 1. Then:*

599 1. **Rank aggregators.** *The composition passes $\{w_i u_i\}$ to an unweighted order statistic over*
 600 *K transformed values per coordinate. The mechanism is preserved when:*

- 601 (i) Count majority: $n_a < K/2$.
- 602 (ii) Signed ordering (C2): *on suppression-relevant coordinate k , the adversarial trans-*
 603 *formed value lies strictly beyond every benign one: $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper*
 604 *tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail).*

605 Sufficient magnitude condition for (ii): *if $\text{sign}(u_{a,k})$ is common to all adversaries and*
 606 *$|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds. This branch is one of two mechanism-preserving*
 607 *regimes, the other (strong attenuation, $w_a \rightarrow 0$) being Lemma 1.*

608 2. **Geometric median** (RFA): *the mechanism is preserved when separation dominates dis-*
 609 *placement: the minimum adversarial–benign separation among the transformed updates*
 610 *strictly exceeds twice the benign spread plus the displacement the adversaries induce in*
 611 *the geometric median (Appendix I defines the four quantities and proves it), under which*
 612 *every adversarial residual exceeds every benign one, so RFA assigns adversarial updates*
 613 *strictly lower inverse-residual weight.*

614 *Proofs are in Appendix I. Both parts are sufficient conditions for d_2 to preserve the downstream*
 615 *suppression ordering, not to exclude adversarial contributions outright; **neither is sufficient for low***
 616 *ASR, and part (2) constrains the aggregate adversarial coefficient only through Corollary 1.*

617 Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share
 618 of the aggregate. A realized *separation margin* $m \geq 1$, the factor by which that condition is met,
 619 defined with the bound in Corollary 1 (Appendix I), does bound it, but only quantitatively: as $m \rightarrow 1$
 620 the bound degrades to the adversarial count fraction, so **the separation condition on its own is not**
 621 **enough**. This is the one place the theory predicts a number and the measurement lands inside it: at
 622 the measured $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds) the bound is
 623 $\Lambda_a \leq 12.6\%$, against a realized $\Lambda_a \leq 1.8\%$. The $m \rightarrow 1$ boundary is where §3’s heterogeneity
 624 failure sits, and it is reached ($m=1.01$ at $\alpha=0.1$).

625 Λ_a is level (P4) in measurable form. (P4) is the first level whose content is the adversary’s
 626 influence on the aggregate, and for a geometric aggregator Λ_a is that level made measurable:
 627 Corollary 1 bounds it, Lemma 1 drives it to a benign-only convex combination, and §4’s instru-
 628 ments intervene on it directly. **So the level the evidence points to is one the theory can bound,**
 629 **just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. Unqualified,
 630 “admission” means *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate
 631 at all; the three aggregator-specific quantities Definition 1 sets out are never used interchangeably
 632 (Appendix G).

633 The remaining case dominates in practice: FoolsGold drives the adversarial weight to zero. This
 634 does *not* remove the adversary from RFA’s input (Definition 2): RFA receives a zero vector, a point
 635 of the geometric-median problem like any other. The correct statement is stronger than removal:

Lemma 1 (Zero-weight annihilation). *Our implementation runs the ε -smoothed Weiszfeld iteration, whose fixed point $\mu' = \sum_i \lambda_i u'_i$, $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$, is a convex combination for every configuration, so no non-degeneracy assumption is needed (Appendix I). If $w_a = 0$ for every adversarial client then $u'_a = 0$ for each, and writing $\Lambda_a := \sum_{a \text{ adv}} \lambda_a$ the adversarial terms vanish identically:*

$$\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}, \quad \bar{u} := \sum_b \frac{\lambda_b}{1 - \Lambda_a} u'_b,$$

*a convex combination of benign transformed updates. The aggregate is a **contraction of a benign-only weighted average**: u_a does not appear in μ' at all, and the origin point's only effect is a smaller effective step: **a utility cost, not a security failure**.*

The same argument covers CoordMedian. The two cases are exhaustive over every measured round (FG zeroes all adversaries in 32% of adversary-present rounds, the separation condition holding in the other 68%), and the mechanism survives for *opposite* reasons: under Lemma 1 the adversaries hold most of the Weiszfeld mass ($\Lambda_a = 0.71$) but carry no payload; under Corollary 1 they carry payload but at most 12.6% of it. The contraction $1 - \Lambda_a = 0.29$ is composition's accuracy cost: FG→RFA reaches accuracy 0.50 and FG→CM 0.66 against 0.78 for the aggregators alone, paid even when the zeroed clients are benign.

The proofs below are short; they are collected here rather than inline because what each result *rules in or out* is the content, not the derivation. Both witnesses of Proposition 2 are runs reported in §4 and §5: Mode S into Krum for part (a), `cos_krum` for part (b).

Proof of Proposition 4. The load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are unoccupied whatever the value of P . A contrast varying d_2 therefore compares pairs differing in P only inside the selected region $M(d_2, a) < \theta$, where P and $M(d_2, a)$ are both functions of the same choice of d_2 ; the missing cell is the one in which they vary independently, and no reweighting of the occupied cells recovers it. The emptiness follows from the gate rather than from $\mathcal{D}$. $\square$

Proof of Lemma 2. C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not attained at d_1 , so it is attained at d_2 . $\square$

Proof of Proposition 5. Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

Proof of Proposition 2. Both halves are exhibited. (a) Mode S into Krum under model-scaling, $\kappa=0$ against $\kappa=2$: ρ moves from 1 to 54.6 and Krum's decision changes in up to 0.80 of rounds, while the support of the admitted adversarial mass changes in none of the 60 measured rounds (a floor on this arm: no adversary is admitted at any rung), the adversarial coefficient share is constant at 0.266667, and suppression shows **no evidence of a practically meaningful change under the pre-specified ± 0.15 equivalence margin** ($|\Delta|=0.026$). (b) `cos_krum` under the pixel backdoor: by Proposition 1(a) its statistic is invariant, and its decision and admission are measured bit-identical (0.000) at every rung, yet ASR falls 0.173 by the first rung (§5). $\square$

Proof of Proposition 3. By construction: Mode S pins every adversarial coefficient at $c=1$ and spreads only benign clients, giving $\Delta_c \equiv 0$ at every ρ ; Mode A holds the benign weights uniform and sweeps only the adversary-to-benign ratio, moving Δ_c at fixed dispersion. Realized values are tabulated in Appendix R, computed with no ASR involved. $\square$

Proof of Proposition 1. (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so the matrix of pairwise cosines is unchanged; FoolsGold's weights are a function of that matrix alone (via each client's normalized maximum similarity), hence unchanged. (c) With $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3 becomes the *closest* point, reversing the ordering Reputation depends on. $\square$

Proof of Theorem 1. (1) CoordMedian operates on $\{w_i u_{i,k}\}_{i=1}^K$ per coordinate. If the adversarial values lie strictly beyond all benign values on one side (ii) and $n_a < K/2$, then at most $n_a < \lceil K/2 \rceil$ of the extreme order statistics are adversarial, so the median is a benign value. For

the magnitude condition: $w_a > 0$ gives $\rho \geq \max_b w_b/w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a common sign this places $w_a u_{a,k}$ strictly beyond every benign value in that direction. Model-scaling sets $u_a = S\tilde{u}_a$ with $S=10$, which amplifies the adversary’s own direction and so satisfies the sign assumption by construction; with measured $\rho \leq 2.1$ the margin is $S/\rho \geq 4.8$. TrimmedMean additionally needs the trimming budget to cover the adversarial count.

(2) Every benign residual satisfies $\|u'_b - \mu'\| \leq R'_B + \delta'$. For adversarial u'_a : $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$, so $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Requiring this to exceed $R'_B + \delta'$ gives (i). $\square$

Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share of the aggregate: several adversaries could each be individually down-weighted yet collectively hold substantial mass. The separation margin controls this, quantitatively:

Corollary 1 (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ be the realized separation margin, so Theorem 1(2) reads $m > 1$. Benign residuals are at most $D_b := R'_B + \delta'$ and adversarial residuals at least $(2m - 1)D_b$, so bounding $\sum_j 1/d_j$ below by the benign terms alone gives*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a}{n_b(2m-1) + n_a}.$$

The aggregate is thus a convex combination whose adversarial share is at most Λ_a : writing $\mu' = (1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for the benign and adversarial parts, the displacement from the benign-only combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.

The bound is only as good as the margin: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, approaching 40% of the mass at $n_a=2, n_b=3$. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds), capping the adversarial share at 12.6% for $n_a=2$; the realized Λ_a never exceeds 1.8%, and the bound holds in all 17 measured rounds.

J Body material, at full length

The body compresses the following; nothing here is a new claim, a new number or a new run. Figure 2 summarizes the preservation chain and the two directions in which it breaks.

J.1 Contributions, organization and scope in full

§1 states these compressed. Nothing here is a new claim.

J.1.1 The three evidence tiers

Table 2: The three tiers of evidence, of very unequal weight and never pooled. §1 states them in prose; every empirical claim in the paper is tagged with one of the three words where it is made.

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§4)	statements about the <i>mechanism</i> , scoped to these cells
Replication	EMNIST-byclass, second dataset <i>and</i> architecture ($\Delta=-0.017$), and <code>coord_median</code> , a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
Screening	the outcome-gated 42-pair suite, 38/42	evaluation prioritization only ; by Proposition 4 it cannot identify the mechanism’s contribution at <i>any</i> sample size

J.1.2 The five contributions

1. **The preservation hierarchy, and the two-directional break above.** Because the per-client defenses in use are all positive scalar rescalings, mechanism preservation is a decidable *invariance* question (Proposition 1), and on the weighting→aggregator class we prove it with explicit

716 margins (Theorem 1), a **boundary** theorem, not a security theorem. Definition 1 gives the five
 717 levels the box states, and §4 measures both breaks. One corollary is worth stating alone: weight
 718 heterogeneity is safe at *both* extremes and dangerous in the middle, so **the weight ratio ρ alone**
 719 **mispredicts** (Lemma 1, §R.1).

720 2. **An identification boundary for outcome-gated screening, and the design class it leaves**
 721 **admissible.** Proposition 4 proves the general claim above; its FL instance is that $C0 \wedge C1$
 722 force the downstream defense of every certified pair to suppress the attack alone, **mechanism-**
 723 **effectiveness confounding** (§3), and its evidential cost is real: a matched ablation built to break
 724 the confound had its premise refuted by its own pre-freeze baselines, so **the confound re-**
 725 **materialized inside the experiment designed to remove it.** A second boundary limits coverage
 726 rather than validation: **emergent** suppression (present in the pair, absent in both constituents)
 727 cannot satisfy an inherited-suppression gate, capping such a screen’s recall (Proposition 5), and
 728 our own best composition is its witness.

729 3. **Three central empirical findings, all causal interventions, not a validation of any screen.** (i)
 730 *The negative, instrumented and replicated (Causal, then Replication).* Mode S pins every ad-
 731 versary’s coefficient, closing Lemma 1’s attenuation channel *by construction*; a **pre-registered**
 732 **replication** in a *different* invariance class and on a *second dataset and architecture* reproduces it
 733 while **refuting the cross-arm admission ordering we had predicted** ($R^2=0.239$ over 26 cells),
 734 so **admission is a direction to measure, not a quantitative predictor.** (ii) *A sign reversal under*
 735 *a change of intervention (Causal).* The same cell yields -0.272 ($p_{\downarrow}=0.0015$) with adversarial
 736 influence free to co-vary and $+0.098$ with it pinned: the confounded design does not lose power,
 737 **it gets the sign wrong**; hence we withdraw the pooled form of C2’s causal claim (80 runs;
 738 $\rho_s = +0.351$, $p = 0.091$; §5). (iii) *Heterogeneity closes the theorem’s own margin.* At Dirichlet
 739 $\alpha=0.1$ RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$, the limit
 740 at which Corollary 1’s bound goes vacuous, and two of three certified compositions cross the
 741 threshold (§3). A fourth predictor falls with them: **adversary-correlated dispersion Δ_c** is in-
 742 dependently controllable from ρ (Proposition 3) and locates the two real anchors our synthetic
 743 curve misses, but **orders suppression no better.** Statistic disturbance, admission magnitude and
 744 coefficient targeting are each refuted *as a predictor*: what survives is a *preservation level*, not a
 745 statistic of the coefficient vector.

746 4. **The screening rule is a consequence of (2), not a claim of its own** (*Screening* throughout).
 747 What survives the negative is a conservative *evaluation-prioritization* rule: read C2 off the down-
 748 stream signal’s invariance class, then evaluate the survivors first. We report it as an application
 749 (§3), never as a validated security predictor; it certifies only *inherited* robustness under a *com-*
 750 *mitted* attack, at recall 40%, and outputs an evaluation *ordering*, never a deployment decision.

751 5. **Secondary results.** Fixed-order composition beats randomized ordering $9\times$, because preser-
 752 vation conditions are order-sensitive; and **one** whitebox composition-aware *adaptive* stress test
 753 raises $FG \rightarrow RFA$ to 0.167 ± 0.069 , $3.7\times$ its seed-matched 0.045 on the pixel arm it is built from,
 754 **an inherited-robustness guarantee is not adaptive robustness** (§6, Appendix O).

755 J.1.3 How the paper is organized, as an argument

756 Composition creates an **identification problem**. §2 defines the object to be identified: mechanism
 757 preservation as invariance, the five levels, the conditions; §3 characterizes the **boundary** on iden-
 758 tifying it and *derives* the class of designs that can; §4 runs the experiment that class dictates; §5
 759 reports what the boundary **falsifies**, our own pooled hypothesis included; §3 states what survives.

760 J.1.4 Why this class, and the scope discipline

761 Everything we *prove* is instantiated for positive per-client rescalings, deliberately: the class is a
 762 **laboratory for studying composition, not a general model of it**, it covers the per-client defenses in
 763 current use (norm clipping, FoolsGold, RFA reweighting, FLTrust) and makes invariance decidable
 764 from definitions. **Every claim here is general, formal or empirical, and they do not transfer**
 765 **upward.** Table 1, first in the appendix, is the one place each scope condition is stated (what falls
 766 outside the class, what the theorems do and do not bound, the sample sizes, what “refuted” means,
 767 why our instruments are not deployable) and we do not restate them at each use.

768 J.1.5 Threat model and scope

769 We consider persistent backdoor attacks where an adversary controlling fraction f of clients embeds
770 triggers that survive across rounds, and define *persistence* γ as the fraction of backdoor ASR retained
771 per round of clean aggregation ($\gamma \approx 0.66$ for CIFAR-10/CNN). Attacks include model-scaling [Bag-
772 dasaryan et al., 2020], pixel-pattern backdoors [Chen et al., 2017], distributed backdoor attacks [Xie
773 et al., 2020], and adaptive composition-aware adversaries [Shejwalkar and Houmansadr, 2021]. The
774 framework constrains the aggregation pipeline, not the communication graph, and is evaluated only
775 in **star-topology FL**. **A scope condition on the criterion itself, not a limitation of our runs:** C1
776 and every prediction built on it are defined against a *committed* adversary, so what is bounded is
777 *inherited* robustness under a fixed attack, **never adaptive robustness** (Table 1).

778 J.2 The prioritization criterion, and where it breaks

779 **This section is a consequence of the boundary, and what survives is an *evaluation-prioritization***
780 ***criterion*, not a security screen.** C1 $\wedge$ C2 answers one question: given a menu, which ordered com-
781 positions are worth running first. **We disclaim predictive validity explicitly;** it does not predict
782 whether a composition is secure, its positive direction is unvalidated, and the cross-arm admission
783 ordering a predictive reading would need was *refuted* by our own pre-registered replication (§4). It
784 also costs us our own best composition: under the per-attack quantifier **FG $\rightarrow$ RFA no longer sat-**
785 **isfies C1**, so the lowest-ASR composition we found is a **false negative** and Proposition 5’s witness
786 (uncertifiable by *any* inherited-suppression screen rather than missed by ours) while the certified in-
787 distribution set (FG $\rightarrow$ CM 0.107, Rep $\rightarrow$ CM 0.423) is correct and the miss costs nothing detectable
788 (0.093 against 0.107, Welch $p=0.52$). Out of sample it licenses *positive* predictions only, against un-
789 informative FAIL controls, so **this is not an out-of-sample classification result:** the *Screening-tier*
790 38/42 at recall 2/5 rests on the in-distribution tiers. The defensible claim is *conservative prioritiza-*
791 *tion*; scoring condition by condition, the confusion matrix, the false negatives and what the rule can
792 and cannot do are in Appendices S–T, U and Table 9 (App. V).

793 **The procedure, and where it breaks.** (0) If one defense already suppresses every attack you
794 have, stop (C0); (1) require a constituent that suppresses the committed attack (C1); (2) read C2 off
795 the downstream signal’s *invariance class*, which §3 says you cannot test directly; (3) read C3 as a
796 margin from a short pilot, never as a gate. Then **evaluate the survivors first, and evaluate them:**
797 none of this certifies security. **Where the rule breaks is measured rather than argued: under**
798 **extreme heterogeneity (Dirichlet $\alpha=0.1$) two of three certified compositions cross the threshold**
799 **and RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$:** the $m \rightarrow 1$
800 limit at which Corollary 1’s bound says nothing. Theorem 1’s guarantee is therefore not merely
801 conservative under heterogeneity but *empty* there, and the theorem earns its keep by predicting its
802 own failure region: a short pilot measuring $m \approx 1$ declines to deploy. **Post-hoc, not pre-registered**
803 (Appendix O.1).

804 J.3 What is proved, what is supported, and the six-step procedure

805 **What is proved, what is supported, and what is not.** **Proved:** the invariance classes (Prop. 1);
806 sufficient conditions for mechanism preservation on the weighting $\rightarrow$ aggregator class (Thm. 1,
807 Lem. 1); the identification boundary and the design class it leaves admissible (Prop. 4, Lem. 2);
808 the recall bound on inherited-suppression screens (Prop. 5). **Empirically supported** (*Causal*, then
809 *Replication*): that substantial disturbance of the downstream statistic need not produce a corre-
810 sponding change in suppression when the adversarial coefficient share is held fixed, on the Krum
811 arm under Mode S and, by pre-registered replication, in a *different* invariance class and on a *second*
812 *dataset and architecture*; and that the criterion behaves as a conservative prioritizer on 42 pairs. **Not**
813 **established:** its positive predictive direction ($R^2=0.239$, cross-arm ordering refuted), and general-
814 ization beyond the scope Table 1 fixes, $\gamma \approx 0.66$ being architecture-conditional. The criterion **trades**
815 **discovery for reliability:** it rejects working compositions (recall 40%, one of them ours), cannot in
816 principle do otherwise (Prop. 5), and draws negative-direction evidence only from the in-distribution
817 tiers. FG $\rightarrow$ CM is an evaluation *ordering*, **not a deployment recommendation** (Appendix H).

818 **What to do instead: six steps for any composition claim.** (1) Name the downstream mechanism
819 and its invariance class. (2) State *which* preservation level you claim, (P1)–(P5). (3) Check the

upstream transform lies in the class your argument covers: C0 here; one altering participation, carrying cross-round state or mixing clients is covered by no invariance argument. (4) Hold d_2 , the attack, and therefore the gate’s input fixed; **do not select candidates on the outcome**. (5) Intervene on the upstream transform and **measure the channel**: aggregate displacement, decision, admission, influence, not the outcome-conditional contrast. (6) Report the level at which the chain broke. Steps (4)–(5) are what Proposition 4 licenses, and following them turned our -0.272 into $+0.098$; **this, not the criterion, is what we ask readers to carry away**. One boundary it does not cross: under one whitebox composition-aware *adaptive* stress test $FG \rightarrow RFA$ rises to $ASR = 0.167 \pm 0.069$, $3.7 \times$ its seed-matched 0.045 on the pixel arm it is built from, so **an inherited-robustness guarantee is not adaptive robustness**: the condition the framework most lacks (Appendix O).

J.4 The identification boundary, formally

This section states §3’s two propositions and their FL instance at full length; the body gives the boundary in words and the design it dictates.

Proposition 4 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2 , a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense**.***

The structure is a collider: the gate E is a function of constituent effectiveness, effectiveness affects the composition’s outcome, and the mechanism condition P is fixed by the same pair of definitions, so conditioning on E makes P and effectiveness co-vary across every certified candidate. The design table is the whole argument, and **the missing counterfactual is a cell, not a sample size**:

	$M(d_2, a) < \theta$ (d_2 suppresses a alone)	$M(d_2, a) \geq \theta$ (it does not)
P holds (mechanism preserved)	observed: every certified pair	empty by the gate
$\neg P$ (mechanism destroyed)	observed: predicted-FAIL pairs	empty by the gate

Reading down a column contrasts P against $\neg P$ at fixed effectiveness (the comparison that would identify P ’s contribution) and the right-hand column, the only one in which d_2 is not already doing the work, is empty for every menu $\mathcal{D}$. **The counterfactual needed is excluded by design, not undersampled**. The emptiness follows from the gate, not the menu, and nothing in the argument is specific to FL, to backdoors, or to our conditions (Appendix I). Its instance for *our* criterion is one line:

Lemma 2 (C0 and C1 place the effective constituent downstream). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $ASR(d_1, a^*) \geq 0.5$. Then $ASR(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Instantiated: in every in-scope pair our conditions certify, d_2 already suppresses the attack d_1 fails, so a certified composition’s low ASR is always attributable to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3): confounded **by the conditions themselves**, not by an unlucky menu. We name this **mechanism–effectiveness confounding**.

This is selection on the outcome; where it hides is the point. The hazard is standard (conditioning on a descendant of both the putative cause and the outcome [Elwert and Winship, 2014, Pearl, 2009]) and **we claim no novelty for it**. What is not standard is that in composition studies the gate is presented not as conditioning but as a *reasonableness* filter (“we study defenses that work”), so it reads as hygiene rather than as the step that removes identification. Two consequences are routinely assumed away: **a better estimator cannot fix it**, no reweighting of occupied cells recovering an unoccupied one, and **the gate cannot be dropped**, because without it C1 is vacuous.

870 **Both worked instances are ours, and one of them inverted a sign.** We name no other paper;
871 our own two designs are the instances. The C1-gated study of §5 is the template at the level of the
872 *criterion*; the dose ladder of §5 is a second instance one level down, its gated quantity being the
873 dose’s incidental attenuation of whichever client carries the poison. **We ran the outcome-gated**
874 **design first, and on the identical cell it returned the opposite sign** (Table 3): the hazard cost us
875 the sign of our own effect, in the direction that flattered us.

876 **The admissible design, derived: this, not the criterion, is the practical contribution.** Read
877 forwards the boundary is *constructive*: each clause below is forced by a term in Proposition 4, not
878 chosen for convenience. **(1) Hold d_2 and the attack fixed**, so the gate’s input $M(d_2, a)$ is numeri-
879 cally the same quantity on both sides of every contrast and cannot co-vary with the mechanism. **(2)**
880 **Intervene on the upstream transform** instead of selecting a different d_2 off a menu, because it is
881 varying d_2 that empties the identifying cell. **(3) Pin the adversarial coefficient channel**, because
882 an upstream that attenuates the adversary moves the outcome by a route with nothing to do with d_2 ’s
883 statistic: the confound that inverted our own sign above. **(4) Measure every channel: aggregate**
884 **displacement, decision, admission, influence, rather than the outcome-conditional contrast**, the
885 outcome being exactly the variable the gate conditioned on. §4 is clause (3) built as an instrument;
886 Appendix J.3 states all four as a six-step procedure for any composition claim.

887 **A second boundary: what such a screen cannot find.** The gate that limits what a screen can
888 be *validated* on also limits what it can *cover*, which converts the most obvious objection to our
889 criterion, that its recall is poor, into a scope result.

890 **Proposition 5** (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$ ’s*
891 *suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent*
892 *composition satisfies the eligibility gate of Proposition 4. Hence, writing S for the compositions*
893 *with $M < \theta$ and $\mathcal{E} \subseteq S$ for the emergent ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|S|$*
894 *on every menu and every attack.*

895 **This is a scope theorem, not a defect found by tuning**, and it is specific neither to C1 nor to
896 FL: any screen requiring a constituent to suppress the attack independently inherits it. Our own
897 lowest-ASR composition is the witness (FG→RFA reaches 0.093 where FoolsGold alone reaches
898 0.732 and RFA alone 0.845), so at $|\mathcal{E}|/|S| = 1/5$ the bound caps any such screen’s recall at 80%,
899 above the 40% we measure (§3). Screening for inherited robustness and searching for synergistic
900 compositions are different problems; only the first is ours.

901 **What the boundary rules out, our own previously-decisive test included.** These proofs are
902 short and their value is what they rule in and out. They rule *out* the experiment we had previously
903 named as decisive: a d_2 whose signal is invariant under positive rescaling but *weak in isolation*
904 cannot satisfy C1 in scope, so no such certified pair exists to test. §4–§5 run four members of
905 the class they rule in. **The strongest disturbs the statistic while pinning the adversarial mass,**
906 **isolating statistic disturbance from adversarial coefficient attenuation, so we built it (§4).**

907 J.5 The targeted suite in full

908 The body compresses these paragraphs; nothing here is a new claim or a new run.

909 **Reputation’s confirmation, in full.** With the adversary’s share pinned, reputation’s rise at $\rho=54.6$
910 shrinks from +0.762 to +0.178 (one-sided Welch $t=-5.22$, $p=0.0017$); what remains tracks a
911 small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision change ($0.000 \rightarrow 1.000$).

912 **The arc that produced this section.** Our pre-registered pooled hypothesis (H1) (ASR should rise
913 with measured disturbance of the statistic the downstream defense reads) was **rejected**, the effect
914 coming out *negative* (-0.272 , $p_{\downarrow}=0.0015$ on the flagship cell), because that ladder moved adver-
915 sarial influence and the statistic *together* and so never tested H1 at all (§5). This section builds the
916 instrument that fixes it, pinning the adversarial coefficient share while the statistic moves, so statistic
917 disturbance is isolated from adversarial coefficient attenuation. C2 is written at level (P2); if losing
918 (P3) meant losing (P5) the criterion would transfer straight to security, and the experiment below
919 adjudicates that implication by **oracle intervention on two arms in two invariance classes** (*Causal*
920 tier), not by a law about defense compositions in general.

An instrument that closes the attenuation channel by construction. A positive rescaling reaches d_2 through two channels at once: it disturbs the *statistic*, and it *attenuates* whichever client carries the poison (Lemma 1). Separating them needs an upstream that moves the first while pinning the second. **Mode S** does this: it fixes every adversary’s coefficient at $c=1.0$ exactly and spreads only the benign clients over $\rho = e^{2\kappa}$, so the **adversarial coefficient share is constant at 0.266667 across all four rungs** (spread 3.5×10^{-7}), and **a rise in ASR here cannot be attenuation**. Design and thresholds were frozen before any run, the share verified with no ASR involved (Appendix X). **Mode A** sweeps the payload instead (§5). **Both are oracle instruments, not defenses and not models:** they read adversary identity, which no deployable defense can, and they do **not** quantitatively reproduce any real defense’s behaviour, at matched disturbance two real upstream anchors sit $+0.767$ and $+0.789$ above the synthetic curve (§5), the measured size of that gap. Their purpose is causal identification, not defense evaluation.

Two readings of C2, frozen against each other. *H-statistic:* what matters is disturbance of d_2 ’s statistic, read off its *decisions*. *H-admission:* what matters is the *adversarial input* d_2 *admits*. On the Krum arm these predict opposite outcomes, since Mode S disturbs Krum’s selection heavily while, the adversary’s weight being pinned, leaving the adversarial mass it admits untouched. That arm adjudicates (Table 4, App. N): as ρ climbs to 54.6, Krum’s **decision changes in up to 0.80 of rounds** while its **admission changes in none of the 60 measured rounds**, and ASR shows **no evidence of the predicted increase, remaining inside the pre-specified equivalence margin** ($p_{\uparrow}=0.70$; $|\Delta|=0.026 < 0.15$). **H-statistic is refuted:** on this arm **substantial statistic and decision disturbance does not produce a corresponding change in suppression once the adversarial coefficient share is held fixed, and the evidence points to admission as the relevant channel in these experiments.**

What the instrument does *not* hold fixed, named in full, and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is **already measured and printed:** the Δ agg. column of Appendix C is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense *emits*, and reads **0.892** here. Appendix D splits that 0.892 into 0.986 of *re-selection* and 0.973 of *rescaling*, with none of the re-selections landing on an adversary. So the claim is sharper: **that displacement, with decision changes in up to 80% of rounds, leaves suppression inside the pre-specified equivalence margin, provided the adversarial coefficient share does not move. What is pinned is pinned exactly:** $c_{\text{adv}}=1.0$ at every rung with maximum deviation $0.00\text{e}+00$ and the share constant to $3.5\text{e}-07$ on CIFAR-10, $3.9\text{e}-07$ on EMNIST-byclass, float32 read-back noise, not slack in the instrument (`measure_admission.py`).

Closing the magnitude channel as well: the pre-registered score-only control. Channel (c) is the one confound Mode S leaves open, so we closed it too. **Score-only Krum** scores on the transformed stack exactly as before and aggregates the *untransformed* selected update, so the dose changes *which* client contributes and nothing about *what* it contributes: Δ dec. is the same 0.733 while Δ agg. is no longer 0.892 but the re-selection component alone, 0.986 relative to $\|u_{s_0}\|$. Rules were frozen at 35788d9 before any score-only ASR existed: primary Δ against the same ± 0.15 margin, where $\Delta > +0.15$ would have said **the published result was partly an artifact of the magnitude channel** and narrowed the central claim. **The outcome is $\Delta = -0.023$ (mean ASR $0.062 \rightarrow 0.039$) against -0.026 uncontrolled on the identical cell, so closing the magnitude channel moves the effect by 0.003;** the refuting threshold 0.212 is nowhere approached and every rung clears the accuracy floor. **The central claim is thus instrumented against two independent confounds**, attenuation and magnitude (row six of Appendix C). The secondary trend test, the headroom asymmetry, the selectors-only scope and the `cos_krum` corollary are in Appendix E.

A pre-registered replication, on a different defense class, that refutes the positive half. The two existing Mode-S arms move together on both predictors, so neither adjudicates. `coord_median` under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the **smallest decision change** (0.482) and the **largest admission-level change** ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise. Thresholds were frozen against the two published rises: confirm

if $\Delta > +0.178$, refute if $\Delta < +0.150$ (Appendix X). **The outcome is $\Delta = +0.098$, so the admission ordering is refuted:** the arm with the largest admission-level change rose *less* than one with a smaller change. Nor does the decision reading predict it, having called the *smallest* rise when $+0.098$ is the middle one; the added cells leave the pooled model unconfirmed ($R^2=0.239$, the share coefficient’s sign failing the pre-specified conjunction). **What replicates is the negative**, on a coordinate-wise order statistic rather than Krum’s selector: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698. Two qualifications: this cell’s identity ASR is 0.443, so equivalence is to the *identity rung*, not a low absolute ASR; and the rise is positive in 5/5 seeds (post-hoc), so the cost is **small and consistent, not zero**. **The instrument earns its keep:** the confounded ladder on this identical cell *falls* 0.272 ($p_{\downarrow}=0.0015$), so closing the attenuation channel **inverts the sign of the effect**.

What the two arms jointly establish. Both halves are witnessed by runs already reported, so what follows formalizes measurements rather than adding a claim. **Statistic-level non-identifiability** (Proposition 2, proved from the exhibited witnesses in Appendix I): within the positive per-client rescaling class there are transform families inducing arbitrarily different statistics and *decisions* at the *same* admitted adversarial mass and the same suppression (Mode S into Krum under model-scaling) and families inducing the same statistic, decision *and* admitted mass at *different* suppression: `cos_krum`. Consequently **no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression**. **That corollary is the paper’s title claim:** no metric defined *solely* on preservation of the downstream statistic can identify attack-suppression equivalence. **Scope:** a non-existence claim *within this class on these defenses and attacks*, by exhibited counterexample: **not** that no metric of any kind can order security. A second proposition sharpens §2’s warning: statistic-level dispersion ρ and adversary-correlated dispersion Δ_c are **independently controllable** ($\Delta_c \equiv 0$ while ρ ranges over $[1, 54.6]$; Δ_c over $[-1.138, +2.469]$ at fixed dispersion), so **a transform reported by ρ alone leaves Δ_c entirely free** (Proposition 3), which does *not* make Δ_c a predictor, its magnitude failing to order suppression either (§5). The instrument is elaborate because §3 leaves nothing simpler. §5 answers what remains: *what happened when we tried it the obvious way first?*

1004 K What the boundary licenses: the suites in full

1005 This section holds §5’s three suites at full length: the two attempts the identification boundary
1006 defeated, the confounded dose ladder that motivated Mode S, and the remaining targeted arms.
1007 Nothing here is rescored; the body states the outcomes.

1008 K.1 Two attempts that the confound defeated

1009 Both were built to validate the *screen*; neither could, so they are evidence for Proposition 4 rather
1010 than for the framework.

1011 **Two pre-registered suites, and the two ways the boundary asserted itself.** (i) *A prospective*
1012 *suite on three unused defenses: 13/20, and no positive test.* Everything scored in §3 was developed
1013 on the menu it is scored on, so we ran a prospective test on FLTrust [Cao et al., 2021], Krum and
1014 Multi-Krum [Blanchard et al., 2017], for which the theory predicts *opposite* outcomes. All 20 labels
1015 were git-committed before the output directory existed and both attacks clear the FedAvg sanity gate,
1016 so the negatives are informative (11/13 in scope, 5/7 on the falsifiable arm; Appendix Q). **Three**
1017 **disclosures dominate it:** FLTrust alone suppresses both attacks (0.048), so its 7 pairs are **out of**
1018 **scope** by C0 (applied *retrospectively*); its 6 in-scope predicted-LOW pairs are **near-identities**, four
1019 bit-identical to FLTrust alone, so **this suite supplies no falsifiable prospective positive test** (one
1020 arrived only with the census of §K.2, and it is a single cell); and MultiKrum at $K=5$ selects all five
1021 participants and so is bit-identical to FedAvg, which together with the pass-through d_1 rows leaves
1022 8 of the 20 rows reducing to a single defense and the falsifiable arm with 2 pairs composing two
1023 mechanisms rather than 7. (ii) *A matched metric-swap ablation, whose premise its own baselines*
1024 *refuted.* To break the confound by construction we held the suppression *principle* fixed and varied
1025 only the invariance class, mirroring Reputation and Krum scoring from *directions* alone. Measuring
1026 the C1 input before freezing **refuted the design’s premise:** `cos_reputation` collapses (0.017 $\rightarrow$
1027 0.982) and `cos_krum` reaches 0.078 only at 0.146 accuracy, failing the C1 gate applied everywhere

1028 else (**magnitude information is what lets this family handle a magnitude attack**), so C1 bound
 1029 for all eight pairs, every label was HIGH, and the suite bears on specificity only (8/8; App. L, P).
 1030 **The confound re-materialized inside the experiment built to remove it.**

1031 **The one contrast that escapes, and the design it dictates.** One comparison evades the boundary:
 1032 hold d_2 and the attack fixed and vary only the upstream, so C1 is identical by construction. On
 1033 *Krum*/model-scaling, NormClip disturbs its selection 0/9 and reaches 0.062 while RFA disturbs it
 1034 6/9 and reaches 0.779 (Welch $p=0.012$, $n=5$), with no overlap across the six cells where standalone
 1035 d_2 genuinely suppresses. **This is post-hoc** (Appendix L) but has what nothing above it does: C1
 1036 held exactly fixed, disturbance measured on the mechanism rather than read off the outcome.

1037 K.2 A third attempt that closed a census, pre-registered

1038 **The region where mechanism preservation is settled on both sides is small enough to enu-**
 1039 **merate, so we censused it rather than sampling it.** It requires a d_1 emitting strictly posi-
 1040 tive per-client coefficients $T(u_i) = c_i u_i$, so Proposition 1(a) applies, against a d_2 reading an
 1041 exactly scale-invariant statistic: over this project’s defenses and the metric-swap mirrors that is
 1042 $\{\text{norm_clip}, \text{rfa}, \text{reputation}\} \times \{\text{foolsgold}, \text{cos_krum}, \text{cos_reputation}\}$, exactly 9 ordered
 1043 pairs, 7 already measured at the identical protocol and 2 unspent. Enumeration is what re-scoped
 1044 a planned 20-pair wave: 118 **ordered pairs of this project are already spent**, and once a non-
 1045 degenerate d_1 and C0 are required only 6 candidates remain anywhere. Pre-registered at 96f8856
 1046 with per-pair C0–C3 verdicts, a frozen scoring rule and a two-sided resolution table; 24 runs, 4.87
 1047 h.

$d_1 \downarrow$ $d_2 \rightarrow$	foolsgold	cos_krum	cos_reputation
norm_clip	0.887 HIGH	0.508 HIGH [†]	0.935 HIGH [†]
rfa	0.914 HIGH	0.566 HIGH	0.861 HIGH
reputation	0.807 HIGH	0.263 LOW*	0.787 HIGH*

Max-committed mean ASR over seeds 42, 43, 44; LOW iff < 0.5 .

* the two unspent cells, labels frozen before the output directory existed.

[†] norm_clip’s clip is inactive on the pixel arm (max norm $3.043 < \tau=5$), so those

pixel cells are bit-identical to d_2 alone; both verdicts rest on the scaling arm, where it is not.

1049 foolsgold is excluded from the d_1 column by measurement, and that measurement changed
 1050 the design before any prediction was frozen. It normalizes with `weights - weights.min()`,
 1051 forcing at least one coefficient to *exactly* zero every round, which puts it outside Proposition 1(a)’s
 1052 strict $c_i > 0$ hypothesis and under Lemma 1 instead: measured through the shipped transform, 2/5
 1053 exact zeros every round and both downstream cosine statistics moved in 9 of 9. Synthetically at
 1054 1000 trials the effect is entirely the zero’s: strictly positive coefficients change cos_krum’s selection
 1055 0/1000 times, one exact zero 471/1000. reputation emits 0/5 zeros at a positive-part ratio up
 1056 to 18839.8, and under that rescaling in the same rounds the cosine statistics moved 0/9 while
 1057 krum’s selection moved 8/9 and reputation’s own ordering 9/9: **a contrast between statistics**
 1058 **on identical inputs, not between conditions.** So foolsgold $\rightarrow$ cos_* were entered as *measured-*
 1059 *C2-fail controls*. **Had C2 been assumed from the definitions rather than measured, this wave**
 1060 **would have reported two mislabelled pairs.**

1061 4/4 as registered, including the pair flagged uncertain. Pair 1 (reputation $\rightarrow$ cos_krum, cer-
 1062 tified) LOW at 0.263; pair 2 (reputation $\rightarrow$ cos_reputation, the C1-fail $\wedge$ C2-true cell that dis-
 1063 criminate the strategies) HIGH at 0.787; the two controls HIGH. The near-identity check is clean
 1064 on all 8 arms (0 of 3 seeds identical to d_2 alone), so these are compositions. Pair 1 was the live
 1065 risk and the prereg said so: its LOW rests on two *different* attacks suppressed by two *different* con-
 1066 stituents, reputation holding model-scaling (0.017 at accuracy 0.777) and cos_krum the pixel
 1067 backdoor (0.300 at 0.598), while on the scaling arm the two nearest measured analogues (0.508,
 1068 0.566) pointed the other way. **So the census’s one low-ASR pair is suppressed by neither con-**
 1069 **stituent alone, and it is the pair C1 $\wedge$ C2 selects:** over the closed census statistic preservation admits
 1070 all 9 at 11% precision for 90 runs, C1 $\wedge$ C2 admits 1 at 100% precision and 100% recall for 10. In
 1071 this region C2 alone is not a screen at all, since everything here satisfies it by construction, and at the
 1072 frozen seed count B dominates A on precision, recall and cost at once: **the opposite of the trade in**
 1073 **§5.** Disclosure (v) below is the qualification that phrase needs.

1074 **Five disclosures.** (i) B’s 100% is the accuracy-gated reading: `cos_krum` alone reaches 0.078 on
1075 model-scaling at accuracy 0.146, barely above CIFAR-10’s 0.10 chance, so it is a destroyed model;
1076 read at the bare 0.5 threshold C1 also admits `norm_clip` $\rightarrow$ `cos_krum` and `rfa` $\rightarrow$ `cos_krum`, both
1077 HIGH, and B selects 3 for 33%. The analyzer emits both readings. (ii) The one positive is $n=3$
1078 and wide: pair 1’s pixel arm is 0.000/0.107/0.681 per seed, so one seed sits above the threshold
1079 and the verdict is the frozen mean rule; we add no seeds after seeing that. (iii) The controls are
1080 weaker than intended: every `foolsgold-as-d1` scaling arm sits at chance accuracy (0.100), and one
1081 control is HIGH only through the accuracy gate (raw ASR 0.086 at accuracy 0.255), so they show
1082 the C2 probe is not inert but are not independent confirmations. (iv) 7 of the 9 cells are retrospective;
1083 only the two `reputation` cells were blind, and the closure is prospective at the margin rather than
1084 throughout. The wave tripped two frozen sanity flags, both reported and neither adjusted: pair 2’s
1085 scaling arm at accuracy 0.790 against the FedAvg reference 0.667, and that same control’s scaling
1086 ASR identically 0.0 across all 3 seeds at 0.100 accuracy. (v) The C1 input that certifies pair 1 is the
1087 lowest-seed reading of a high-variance cell: `cos_krum` alone measures 0.300 on the pixel arm as the
1088 mean over the frozen seeds 42–44 (0.003/0.269/0.630), and the identical cell, same protocol and
1089 identity upstream, is measured twice more in this paper at 0.490 over 5 seeds and 0.584 over 8, the
1090 latter being the value §J.5 reports as `cos_krum` never suppressing. Six of the eight seeds are above
1091 0.5, and at $n=8$ C1 fails for pair 1, B certifies nothing here, and the census’s one low-ASR pair
1092 becomes another false negative; 0.490 is still below the threshold, so C1 survives at $n=5$. We report
1093 this instead of rescoring, because the pre-registration fixes the seed set at three and forbids adding
1094 seeds afterwards; what it costs is the *dominance* claim above, which holds at the frozen seeds and is
1095 not established beyond them.

1096 **K.3 Why the screen has no second dataset: C1’s pixel-arm input is absent on** 1097 **EMNIST-byclass**

1098 **A second-dataset composition wave was designed, costed at ~ 55 h, and then not run, be-**
1099 **cause the quantity it needed does not exist.** C1 is defined on *standalone* suppression, so its
1100 inputs are measured before any per-pair label is frozen: on CIFAR-10 a pair clears it because the
1101 two constituents cover *different* attacks, as in the census above where `reputation` holds model-
1102 scaling (0.017 at accuracy 0.777) and `cos_krum` the pixel backdoor (0.300 at 0.598). On EMNIST-
1103 byclass/simple_cnn the pixel side is missing. Re-measured through the identical runner at the iden-
1104 tical protocol ($N=10$, $K=5$, 50 rounds, $f=0.2$, Dirichlet 0.5, seeds 42/43/44), `coord_median`
1105 reaches **0.609** at clean accuracy 0.836 (per seed 0.717/0.493/0.618), and over the **seven** de-
1106 fenses with an EMNIST-byclass pixel cell that is the *minimum*: `krum` 0.660, then trimmed mean
1107 0.988, `norm_clip` 0.996, `fedavg` 0.997, `multi_krum` 0.997, `rfa` 0.998 (six *distinct* aggregators:
1108 `multi_krum` reduces to `fedavg` at $K=5$, which is why those two agree). **So no pair drawn from**
1109 **them satisfies C1 there:** the screen certifies nothing over them, every such pair is predicted HIGH,
1110 and the strategy comparison of §K.2 cannot be made on this dataset from these inputs.

1111 **The failure is the pixel arm’s, and we do not escape it by dropping the other.** On the model-
1112 scaling arm `krum` does suppress there (0.027 at clean accuracy 0.767), so a scaling-only protocol
1113 would supply a C1 input and manufacture the comparison; we decline it for the reason the prospec-
1114 tive suite declines the mirror move, that the committed protocol requires *each* attack and choosing
1115 the arm that turns out to have an input is attack-shopping regardless of the resulting labels.

1116 **Two things this is not.** It is *not a failed replication*: no prediction was made and refuted, and the
1117 question a screen answers does not arise where nothing suppresses the attack under test, so it is a
1118 scope limit on C1’s precondition. And it is *not unsatisfiability over the whole menu*: `foolsgold`,
1119 `reputation` and the cosine mirrors have no EMNIST-byclass pixel cell, so the coverage is ex-
1120 actly those cells and we state the gap instead of generalizing past it. We report the obstacle
1121 rather than lower f , subset the 62 classes or lengthen training until a positive appeared, each of
1122 which is a new design axis chosen after seeing the obstacle. Finally, the `coord_median` figure
1123 came back *bit-identical* to the corresponding cell of the earlier payoff matrix (0.6091787658... at
1124 0.8363207047...), which by the standard we apply to ourselves in §J.5 establishes protocol *identity*
1125 and not corroboration: the same computation run twice cannot corroborate itself. What it does buy is
1126 that the matrix’s other cells are protocol-comparable, so the six borrowed rows above did not cost an-
1127 other 21 h (`measure_femnist_c1_inputs.py`, `results/femnist_c1_inputs/summary.json`).

1128 K.4 H1, and the confounded design that motivated the instrument

1129 The prospective suite and the metric-swap ablation both tried to validate C2 by *choosing a different*
 1130 *downstream defense*: the move Proposition 4 rules out. The admissible alternative varies only the
 1131 *dose* at fixed d_2 and attack, so the C1 input is the same quantity at every rung and the confound is
 1132 **impossible by construction** rather than matched away.

1133 **The instrument, the abscissa, and the predictions frozen against them.** The synthetic d_1 has
 1134 one parameter, the *dispersion* of a positive per-client rescaling: $c_j \propto \exp(\kappa(2j/(K-1) - 1))$ at
 1135 position j of a (seed, round)-keyed permutation, normalized so $\text{mean}_j(c_j) = 1$. It is Proposition 1's
 1136 form, the **dose is** $\rho = e^{2\kappa}$, $\kappa=0$ is bit-identical to d_2 standalone, and normalizing the mean while
 1137 keying the permutation on (seed, round) holds aggregate scale fixed and stops the dose parking ad-
 1138 versaries at a fixed end: neither of which, as it turns out, pins Δ_c . Frozen grid $\kappa \in \{0, 0.5, 1.0, 2.0\}$,
 1139 four cells across three invariance classes. What matters to C2 is not ρ but what the rescaling does to
 1140 the statistic d_2 reads, so we measured that first, with no ASR involved, establishing that *there is a*
 1141 *dose* (0 at identity, rising to 0.87/0.93 for class (c)) and that *class-(a) invariance holds in code*. Pri-
 1142 *mary, pooled*: across all 16 (arm, κ) cells the ASR rise over that arm's own $\kappa=0$ rung is positively
 1143 rank-correlated with measured disturbance (Spearman, one-sided, $\alpha=0.05$). *Secondary, per arm*:
 1144 monotone rise for class (c), **FLAT** (the frozen label, scored as equivalence) for class (a), shallower
 1145 for class (b); one-sided Jonckheere–Terpstra at $n=5$, “flat” scored as *equivalence* ($|\Delta| < 0.15$, every
 1146 rung below 0.5), a margin fixed in advance. **The ladder was falsifiable both ways**: `cos_krum`'s
 1147 aggregate is still $c_j u_j$ for the selected j , so suppression can be lost through the magnitude channel,
 1148 and a class-(c) arm inside the margin would refute the central fragility claim. **No frozen prediction**
 1149 **uses S** : measured, S is **not a stable per-attack constant** (medians 0.57/0.92, range to 12.59), a
 1150 limit on the theorem's checkability. Design, grid, arms, shapes and refutation criteria were frozen in
 1151 advance (Appendix M).

1152 **Outcome: one arm confirmed, two refuted, one indeterminate; the pooled test fails.** All 80
 1153 runs completed and were scored per seed against the frozen rules (Table 5, App. N). The **primary**
 1154 pooled prediction is **refuted**: Spearman $\rho_s = +0.351$ over 16 cells, one-sided $p = 0.091$, against
 1155 a frozen $\rho_s > 0$ at $p < 0.05$; **in three of four arms ASR falls as the dose rises**. Arm 2 confirms
 1156 decisively: reputation $0.017 \rightarrow 0.779$ ($p = 1.7 \times 10^{-6}$) at nearly constant accuracy, the paper's
 1157 only prospective positive for C2, and it establishes the *destruction* half only. Arm 1 refutes more
 1158 than its own shape: Krum, whose statistic the fragility claim is stated over, shows no rise (inside
 1159 the equivalence margin) despite measured disturbance 0.867, and does not reproduce our within-
 1160 defense contrast ($0.062 \rightarrow 0.779$ under RFA at a *lower* disturbance), so selection-flip rate cannot be
 1161 its mechanism. Arm 3 is indeterminate and uninterpretable anyway (`cos_krum/pixel` falls $0.490 \rightarrow$
 1162 0.086 , the last rung below the accuracy gate) and arm 4 is refuted outright (`coord_median/pixel`
 1163 falling $0.443 \rightarrow 0.170$ at accuracy ≥ 0.69). Both fell where a rise was frozen, which is why neither
 1164 is scored in our favour.

1165 **Two faults, both ours, and now measured rather than conjectured.** *The instrument.* Normal-
 1166 izing $\text{mean}(c)=1$ fixes the aggregate's scale but not any client's *relative* weight, and because the
 1167 dose ignores adversary status a $55 \times$ spread mostly *dilutes* whichever client carries the poison: on
 1168 the same rounds the ladder's Δ_c drifts monotonically negative with the dose ($0.000 \rightarrow -0.140 \rightarrow$
 1169 $-0.240 \rightarrow -0.336$), so three arms measure attenuation rather than C2 disturbance. **The adversary-**
 1170 **targeted dose is the fix**: Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to
 1171 54.6) and Mode A the Δ_c -**swept** one. *The abscissa.* At matched disturbance two of six *real* upstream
 1172 anchors miss the synthetic curve badly (RFA $\rightarrow$ Krum $+0.767$, NormClip $\rightarrow$ Reputation $+0.789$), and
 1173 not through a range mismatch; Δ_c locates the discrepancy, as Proposition 3 says it may, those two
 1174 being targeted far beyond any ladder rung ($-1.21, -0.72$ against -0.34) at ρ of only 14.8 and 4.19
 1175 against the ladder's 54.6: **matched on the statistic, badly unmatched on the targeting. But this**
 1176 **rescues no predictor**: Δ_c 's magnitude does not order suppression either, and FoolsGold's is *pos-*
 1177 *itive*: **post-hoc and exploratory**, computed with no ASR and rescoring nothing (Appendices N,
 1178 R). Two checks did pass: the $\kappa=0$ rung reproduces every standalone baseline exactly per seed, and
 1179 `cos_krum`'s selection never moved (0/120).

1180 K.5 Mode A, the four aggregators, and the pre-registered second dataset

1181 **Mode A: the two regimes, and where they hold.** Mode A (payload-only) closes the other chan-
 1182 nel: it keeps the benign weights uniform and sets the adversary-to-benign weight ratio to $r = e^\nu$
 1183 at fixed mean, so the adversarial share sweeps $0.049 \rightarrow 0.710$ with the statistic’s dispersion held
 1184 fixed: the mirror of Mode S. That gives Theorem 1 a single falsifiable shape: ASR single-peaked
 1185 at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified
 1186 one is discarded by a rank aggregator (Regime A). It is **confirmed for coord_median**, the rank
 1187 aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls
 1188 $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_\downarrow=3.0 \times 10^{-4}$, 6.4×10^{-4}) at accuracy
 1189 ≥ 0.74 (Figure 3). It is **refuted for reputation**, which is monotone, maximal at the most attenuated
 1190 rung (0.894 at $r=0.135$), because an attenuated adversary reads as *consistent* to a reputation score.
 1191 Reputation is not a rank aggregator, so the shape was never licensed for it.

1192 **The same channels, across four aggregators of three structural kinds.** Across selectors on pair-
 1193 wise and on cosine distance (krum, cos_krum), a coordinate-wise order statistic (coord_median)
 1194 and a weighted averager with cross-round state (reputation), the negative is **not Krum-specific**: the
 1195 decision channel moves by 0.482–1.000 on three of the four while **the support of the admitted ad-**
 1196 **versarial mass changes in none of the 60 measured rounds of any of the four**, and on reputation
 1197 and coord_median that mass is *present* at baseline in 48 of those 60, so their counts are not floors
 1198 the way Krum’s is, and the only two arms whose ASR *rises* are the only two with nonzero $\Delta\Lambda_a$. **An**
 1199 **analysis of runs already reported, not a new experiment, and it orders nothing**: the larger $\Delta\Lambda_a$
 1200 (0.033) belongs to the arm with the *smaller* rise (Appendix C).

1201 **A second dataset and architecture, pre-registered.** The same cell on EMNIST-
 1202 byclass/simple_cnn in place of CIFAR-10/cifar_cnn, frozen in git before any EMNIST-byclass
 1203 ASR existed. Premise measured first, no ASR anywhere: the decision moves on 0.643–0.714 of
 1204 rounds, the support of the admitted mass unchanged in all 56 measured rounds (again a floor: no
 1205 adversary admitted at any rung). **Suppression shows no evidence of a practically meaningful**
 1206 **change**: $\Delta = -0.017$ against the frozen ± 0.15 margin, beside CIFAR-10’s -0.026 , at accuracy
 1207 $0.767 \rightarrow 0.835$, so the gate never binds. **Replicated on a second dataset and architecture**: the
 1208 frozen wording, never “generalizes”. Two limits from the freeze: the Jonckheere–Terpstra was
 1209 specified to catch a monotone *rise* and the trend is a *fall* ($z=-1.563$), so it is not read as support for
 1210 flatness; and with the identity rung at 0.027 the refuting branch was reachable (> 0.177) while the
 1211 indeterminate one was not, so equivalence is to the identity rung, not a low absolute ASR.

1212 It addresses neither recall nor discovery, and nothing in §5 is rescored.

1213 L Verification of the invariance mirrors

1214 All eight metric-swap labels were frozen in advance, and these mirrors are mechanism-
 1215 isolating ablations on the same footing as the instruments of §4. We check the statis-
 1216 tic directly rather than inferring invariance from an outcome (verify_cos_invariance.py,
 1217 results/cos_invariance_check.json), comparing each d_2 ’s internal state on $\{u_i\}$ against
 1218 $\{c_i u_i\}$ for the c_i NormClip and RFA actually produce. Over 18 live rounds (3 seeds $\times$ 2 up-
 1219 streams $\times$ 3 rounds) cos_reputation’s weight vector matches to $\leq 2.5 \times 10^{-7}$ with its ordering
 1220 never changing (0/18) and cos_krum selects the same client in 18/18; the Euclidean originals do
 1221 not (Reputation’s weights move by up to 0.222, ordering changes 12/18; Krum’s selection changes
 1222 6/18). Disturbance tracks the realized weight ratio ρ as Proposition 1 predicts: under NormClip
 1223 ($\rho \in [1.00, 4.90]$) Reputation changes in 3/9 and Krum in 0/9, under RFA ($\rho \in [2.11, 17.38]$)
 1224 in 9/9 and 6/9. On 1000 synthetic draws the separation is cleaner still (0/1000 vs. 992/1000
 1225 for the Reputations, 0/1000 vs. 618/1000 for the Krums), and the mirrors match the shipped im-
 1226 plementations they stand in for to $\approx 10^{-8}$ relative aggregate error. **The post-hoc within-defense**
 1227 **contrast of §5 re-uses this frozen suite**, so it is not pre-registered, and two seeds were added after
 1228 seeing the $n=3$ result: a top-up that cut both ways, degrading the invariant control’s Δ from ≈ 0
 1229 to $+0.19$ while leaving the disturbed/undisturbed split without overlap. One asymmetry matters:
 1230 cos_reputation’s *weights* are exactly invariant but its aggregate is not, since the weighted aver-
 1231 age is over the rescaled $c_i u_i$: a magnitude contraction of the Lemma 1 kind, so the prediction is
 1232 two-sided, with ASR staying low while clean accuracy pays a contraction cost.

M Dose–response caveats recorded before the freeze

The pre-freeze disturbance measurement. The abscissa was measured over 5 seeds $\times$ 3 rounds per cell with no ASR involved, which is why it precedes the freeze: the class-(a) mirror’s cosine scores move by at most 1.1×10^{-4} in 0/120 cells, i.e. the invariance holds in the shipped code and not merely on paper, while the class-(c) arms reach 0.87 and 0.93. **Attack assignment.** The class-(c) arms are on model-scaling and the class-(a)/(b) arms on the pixel backdoor: not a free choice, since on model-scaling *no* invariant d_2 suppresses at usable accuracy (§5). “Same attack” is required *within* each ladder, which is where the causal claim lives; across arms we compare curve shapes and slopes, never levels. `coord.median/pixel` sits only 0.057 below threshold, so its binary above/below-0.5 reading is weak either way. The dose is uncorrelated with adversary status by construction but not adversary-*adversarial*; a dose that deliberately up-weights adversaries would be strictly harder and is not run. A synthetic transform is not a real defense, so we overlay the two *real* upstreams already measured on these cells (NormClip, RFA) as anchors at their own measured disturbance; landing far off the synthetic curve would be a finding about the instrument and reported as one. Design, grid, arms, shapes, margin and refutation criteria are frozen in `pre_registration_dose_response.md` at commit e711a95, timestamped before `results/dose_response/` existed and reproduced verbatim in the runner’s arm table. **What it does not address:** recall, and discovery: `fg→rfa` remains a false negative of our own rule.

N Dose–response ladder: per-seed detail, intervals, and the anchor residuals

Table 3: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose, and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap 0.00e+00, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention, which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table (the adversarial share of selected coordinates, the same field in the same artifact) and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. A two-column contrast on one arm: it locates where the two designs differ and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	<code>coord.median</code>	<code>coord.median</code>
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ -0.336	0 by construction (read back 9e–08)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	-0.272 ($p_{\downarrow} = 0.0015$)	+0.098 ($p_{\uparrow} = 0.069$)

Table 6 is every cell of the ladder; Figure 4 plots it on the measured axis and Figure 5 on the nominal one. Both are produced by `analyze_dose_response.py` and `dose_response_figure.py` from `results/dose_response/`; nothing below is transcribed from a log.

Table 4: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`): mean ASR with the two measured channels. **The decision channel moves far more than ASR throughout.** [†] below the 0.35 accuracy gate. [‡] the replication arm, rank-discordant by design, frozen separately: $\Delta=+0.098$ is below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates. Verdicts otherwise scored against readings frozen in advance; per-seed values in Appendix N.

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8\times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 1 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ < $+0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

Table 5: The ladder as measured, $n=5$, per-attack mean ASR (mean clean accuracy). *Predicted* shapes were frozen before any run (Appendix X); $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability of an increasing trend, the only direction any frozen rule tests. An arm meeting neither its confirmation nor its refutation criterion is reported *indeterminate* rather than scored in our favour. [†] below the 0.35 accuracy gate: a collapsed model, not suppression. Intervals, per-seed values, anchor residuals and both plotted axes: Appendix N.

arm (d_2 / attack)	predicted shape	mean ASR (clean acc.)				JT $p_{\uparrow}$	verdict
		$\kappa=0$	0.5	1.0	2.0		
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63	refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6}	confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999	indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998	refuted

1255 **The replication arm’s post-hoc seed-matched difference.** The frozen primary for the
1256 `coord_median` replication (§4) is the *unpaired* $\Delta = +0.098$, and it is not superseded by what
1257 follows. Between-seed variance on this cell is large (identity ASR spread 0.287–0.677), which the
1258 unpaired test cannot see through, so we also report the seed-matched difference: positive in 5/5
1259 seeds, paired $t = 3.39$, one-sided $p = 0.014$. **This is post-hoc and was not pre-registered;** we
1260 report it because it is the honest description of a small consistent effect, and we label it post-hoc
1261 wherever it appears. It does not change the frozen verdict: the admission *ordering* is refuted either
1262 way.

1263 **The primary test, and two post-hoc sensitivities that cannot change its verdict.** The frozen
1264 primary test is a one-sided Spearman correlation of *rise* against *measured disturbance* pooled over
1265 all 16 cells, confirmed only at $\rho_s > 0$ with $p < 0.05$: we obtain $\rho_s = +0.351$, $p = 0.091$, so it
1266 is **refuted**. Two sensitivities are recorded because they are informative about *why*, and are labelled
1267 post-hoc because neither was frozen. (i) The four $\kappa=0$ cells are constructed (0, 0) points (zero
1268 disturbance and zero rise by definition) and dropping them gives $\rho_s = +0.780$, $p = 0.001$ ($n=12$);
1269 so the pooled correlation is carried by the arm-2 rise and destroyed by the two arms whose ASR
1270 falls, not by the identity anchors. (ii) Against the *nominal* dose κ (i.e. $\log \rho$) rather than measured
1271 disturbance, $\rho_s = -0.354$, $p = 0.911$: the wrong sign, which is the same fact as three arms falling.
1272 Neither is scored.

1273 **Per-arm trend tests.** One-sided Jonckheere–Terpstra across the four ordered rungs, 5 seeds each,
1274 $\alpha = 0.05$; the normal-approximation p agrees with a 20,000-draw permutation p to three decimals
1275 throughout. Arm 1: $J = 70$, $z = -0.34$, $p_{\uparrow} = 0.632$. Arm 2: $J = 144$, $z = +4.64$, $p_{\uparrow} =$
1276 1.7×10^{-6} . Arm 3: $J = 29$, $z = -3.10$, $p_{\uparrow} = 0.999$ ($p_{\downarrow} = 9.8\times 10^{-4}$); dropping the gated $\kappa=2.0$
1277 rung leaves $z = -2.06$, $p_{\downarrow} = 0.020$, so the decrease is not an artifact of the collapsed cell alone.

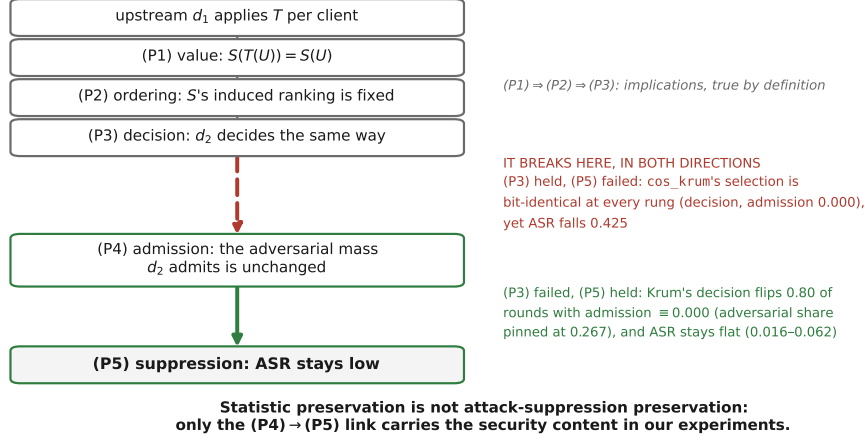


Figure 2: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **below (P3) the chain breaks, in both directions**, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 1’s attenuation channel by *construction* (§4). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder (§5), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering**, so **admission is more causally proximal than the statistic, and not a validated predictor**. Every number in the figure is recomputed from the scoring scripts’ own JSON by `story_chain.figure.py`.

Table 6: All 16 rungs, $n=5$. *disturb* is the independently measured disturbance of that arm’s own statistic at that rung, computed before the freeze with no ASR involved; *rise* is ASR relative to the arm’s own $\kappa=0$ rung and is the primary test’s ordinate. CIs are Student- t at $n=5$ and are wide by design: the ladder was powered to detect a monotone trend, not to resolve single cells. $\dagger$ fails the 0.35 clean-accuracy gate.

arm	κ	ρ	dist.	ASR	95% CI	rise	acc	per-seed ASR (42–46)				
krum scaling class (c)	0	1.00	0.000	0.062	[0.019, 0.105]	+0.000	0.565	0.115	0.021	0.047	0.057	0.069
	0.5	2.72	0.600	0.011	[−0.001, 0.023]	−0.051	0.652	0.000	0.010	0.011	0.027	0.007
	1.0	7.39	0.733	0.014	[0.003, 0.024]	−0.048	0.682	0.003	0.015	0.021	0.021	0.008
	2.0	54.60	0.867	0.054	[0.002, 0.106]	−0.008	0.619	0.005	0.039	0.077	0.113	0.037
reputation scaling class (c)	0	1.00	0.000	0.017	[0.012, 0.022]	+0.000	0.777	0.014	0.012	0.017	0.023	0.019
	0.5	2.72	0.867	0.022	[0.014, 0.031]	+0.005	0.779	0.016	0.018	0.024	0.033	0.020
	1.0	7.39	0.867	0.487	[0.349, 0.624]	+0.470	0.765	0.439	0.547	0.578	0.557	0.313
	2.0	54.60	0.933	0.779	[0.673, 0.885]	+0.762	0.701	0.710	0.843	0.730	0.896	0.714
cos_krum pixel class (a)	0	1.00	0.000	0.490	[0.058, 0.922]	+0.000	0.607	0.003	0.269	0.630	0.864	0.686
	0.5	2.72	0.000	0.450	[0.130, 0.769]	−0.041	0.569	0.764	0.236	0.165	0.648	0.435
	1.0	7.39	0.000	0.153	[−0.127, 0.433]	−0.337	0.466	0.132	0.090	0.000	0.543	0.001
	2.0	54.60	0.000	0.086	[−0.100, 0.271]	−0.405	0.220 [†]	0.083	0.000	0.000	0.345	0.000
coord_median pixel class (b)	0	1.00	0.000	0.443	[0.263, 0.622]	+0.000	0.767	0.395	0.287	0.449	0.677	0.405
	0.5	2.72	0.251	0.428	[0.241, 0.615]	−0.015	0.764	0.454	0.215	0.430	0.638	0.402
	1.0	7.39	0.378	0.327	[0.181, 0.472]	−0.116	0.749	0.351	0.139	0.352	0.463	0.328
	2.0	54.60	0.487	0.170	[0.113, 0.228]	−0.272	0.690	0.179	0.090	0.204	0.192	0.187

1278 Arm 4: $J = 31$, $z = -2.96$, $p_{\uparrow} = 0.998$ ($p_{\downarrow} = 1.5 \times 10^{-3}$). No frozen rule tests the decreasing
1279 direction; $p_{\downarrow}$ is descriptive.

1280 **Clean accuracy against dose separates two ways of losing ASR.** Descriptive, not pre-registered.
1281 Accuracy trends by the same test: arm 1 $z = +1.35$, arm 2 $z = -3.30$, arm 3 $z = -3.50$, arm
1282 4 $z = -4.11$. Arm 3 pays $0.607 \rightarrow 0.220$: a 0.39 collapse through the Lemma 1 contraction
1283 channel, and the reason its $\kappa=2.0$ cell is uninterpretable rather than a success. Arm 4 pays only
1284 $0.767 \rightarrow 0.690$ while ASR more than halves, so there the dose genuinely *helps* coordinate median:
1285 a refutation of our predicted shape, but a benign mechanism. Arm 2 is the cleanest cell in the ladder,
1286 losing 0.076 accuracy while ASR rises 46 $\times$.

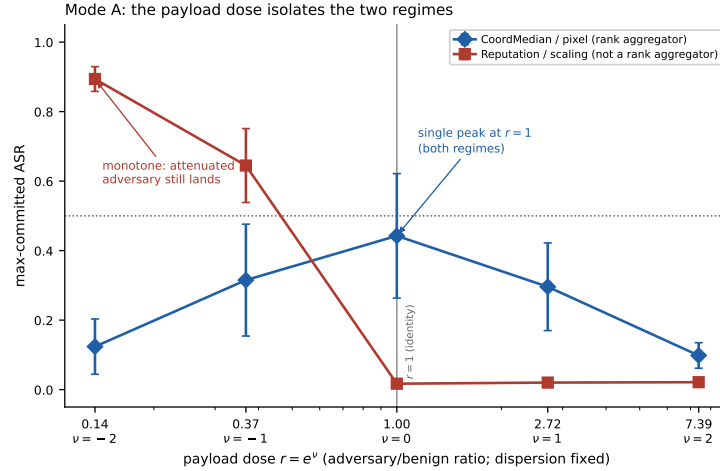


Figure 3: **The two regimes, exhibited: the payload dose separates them.** *The picture of regimes A and B of §2, produced by the instrument of §5.* With the statistic's dispersion fixed and only the adversary/benign weight ratio r swept, `coord_median` (a rank aggregator, the class the regimes are stated for) is single-peaked at $r=1$ as Theorem 1 predicts, while reputation is monotone: off-spec, and the theorem is not claimed for it. Bars are Student- t 95% intervals, recomputed per seed by `analyze_targeted_dose.py`; the Mode-S companion is Table 4.

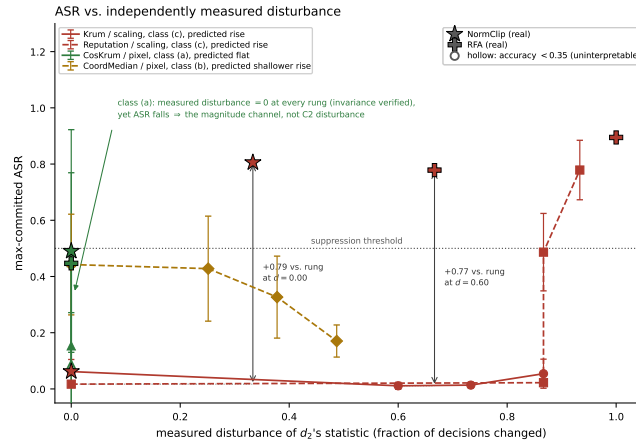


Figure 4: **The measured axis: ASR against disturbance of each arm's own statistic**, an abscissa fixed before the freeze ($n=5$, Student- t 95% intervals; the figure script imports the arm and anchor construction from the scoring script, so the two cannot diverge). Stars and crosses are the two *real* upstream transforms measured on these same cells: at matched disturbance they sit $+0.77$ and $+0.79$ ASR *above* the synthetic curve, though their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) lie *inside* the ladder's span; §5 attributes the gap to Δ_c , post-hoc. The class-(a) arm's four rungs coincide at disturbance exactly 0 and its ASR falls anyway: Lemma 1's magnitude channel, not C2 disturbance. Hollow markers fail the 0.35 accuracy gate.

1287 **Anchor residuals, and why the instrument is the finding.** Residual is real-transform ASR
1288 ($n=5$, from `metric_swap_topup`) minus the synthetic rung at the nearest measured distur-
1289 bance. NormClip→Krum: disturbance 0/9, ASR 0.062 vs. rung 0.062, residual $+0.001$.
1290 RFA→Krum: $6/9 = 0.667$, ASR 0.779 vs. rung 0.011 at disturbance 0.600, residual
1291 $+0.767$. NormClip→Reputation: $3/9 = 0.333$, ASR 0.806 vs. rung 0.017, residual $+0.789$.
1292 RFA→Reputation: $9/9$, 0.895 vs. 0.779, residual $+0.116$. Both `cos_krum` anchors sit at distur-
1293 bance 0 and land $(+0.000, -0.044)$. `coord_median` was not in the metric-swap menu, so arm 4 has
1294 no real-transform anchor and its ecological validity is untested. The two large residuals are not a

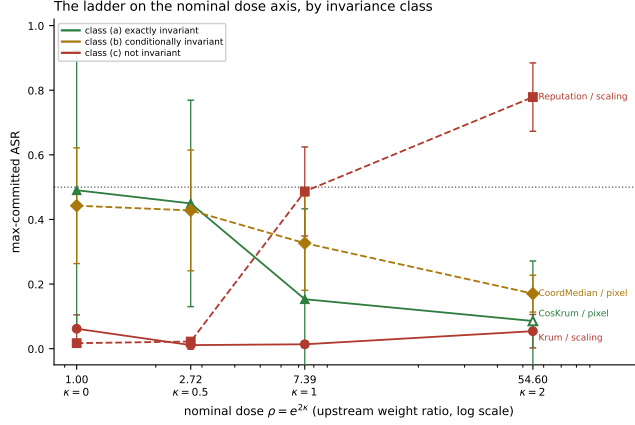


Figure 5: **The nominal axis:** the same 16 cells against the dose $\rho = e^{2\kappa}$, coloured by Proposition 1 class. Of the two class-(c) arms the fragility claim is stated over, only reputation rises; the other two arms *fall*. This is the axis the frozen per-arm shapes were stated on, and the pooled prediction was stated on the measured axis of Figure 4. Hollow markers fail the 0.35 accuracy gate.

range mismatch: realized ρ is mean 3.61 / max 4.90 for NormClip and mean 11.93 / max 17.38 for RFA, both inside the ladder’s $[1.00, 54.60]$ span. What differs is *which* clients get reweighted (NormClip and RFA reweight in a way correlated with adversary status, our permutation deliberately does not) and we state that as the hypothesis caveat (iii) was written to test, not as a measured quantity: the shipped disturbance logs record only $c_{\min}$, $c_{\max}$ and ρ per round, not per-client coefficients, so an adversary-versus-benign coefficient contrast requires a new (cheap, no-ASR) measurement.

Two checks that passed. The $\kappa=0$ rung reproduces each arm’s published standalone value exactly on every shared seed (16 seed-level comparisons across four arms, largest $|\Delta\text{ASR}|$ and $|\Delta\text{acc}|$ both $0.00\text{e}+00$ at tolerance 10^{-6}), so $\kappa=0$ really is the identity and the fresh rung and the published baseline are the same quantity. **The first-rung fall we rest the (P4) $\nrightarrow$ (P5) witness on**, in full: `cos_krum / pixel` at $\rho=2.72$ falls 0.173 with clean accuracy moving only $0.599 \rightarrow 0.567$, paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds. And `cos_krum`’s selection never changed in any of the 120 measured cells, with maximum relative score drift 1.1×10^{-4} : Proposition 1’s classification of which statistics a positive per-client rescaling can disturb is verified directly, independently of the outcome the ladder then failed to predict.

O Secondary results: baselines, the adaptive stress test, scale, and switching

The margin behaves like a certificate, not just a switch. C3 is binary, but the margin m is measurable and its *strength* tracks security: over the 29 measured rounds with $w_a > 0$, higher margin goes with lower realized adversarial mass (Spearman $\rho = -0.48$, $p = 0.009$), Corollary 1’s bound holding in every one, a number rather than a verdict, though the link is directional, not fitted, and post-hoc.

Single-defense baselines. Measured under the identical protocol (5 seeds, 50 rounds), no individual defense achieves low max-committed ASR: CM 0.52, FG 0.73, Rep 0.84, TM 0.85, RFA 0.89, NC 0.94. Against the best single defense (CM, 0.52), FG→RFA (0.093) is $5.6 \times$ lower and FG→CM (0.107) is $4.9 \times$ lower. FoolsGold alone reaches ASR 0.20 on model-scaling only by collapsing the model to 10% accuracy, so it is not suppression: consistent with its C1 failure.

Adaptive attacks against FG→RFA. We evaluate one whitebox adaptive attack family that jointly evades both defenses: disjoint coordinate blocks minimize cosine similarity (evade FG) while ϵ -ball projection stays near the geometric median (evade RFA). The adversary has full state access, and these constraints are in tension. ASR against FG→RFA (5 seeds each, base quoted on the *same* five seeds and the *same pixel attack the adaptive family is built from*, so the comparison is seed-matched and same-attack rather than against the $n=30$ mean or the pairs worse scaling arm): base

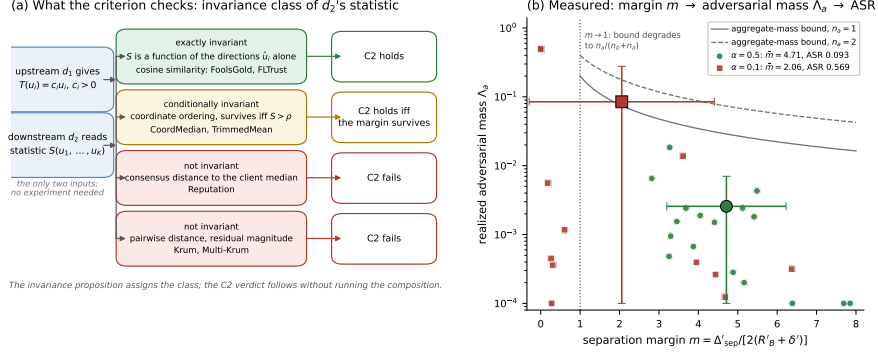


Figure 6: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 1), and the class determines the verdict with no composition run, which is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives; each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 1 and carry no payload), large markers are mean $\pm$ s.d., and values below 10^{-4} are drawn at 10^{-4} . Corollary 1’s bound holds in every round, and the $\alpha=0.1$ collapse is visible in the theorem’s own quantity: m falls from 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a/(n_b+n_a)$, and realized Λ_a rises $0.003 \rightarrow 0.085$ (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$.

Table 7: Framework validation (representative pairs; full tallies in text). Threshold: LOW < 0.5 , HIGH ≥ 0.5 .

Composition	Category	Predicted	Max ASR	Correct?
<i>PASS: C1∧C2∧C3 (predicted LOW)</i>				
FoolsGold → RFA	PASS	LOW	0.093	✓
FoolsGold → CoordMedian	PASS	LOW	0.107	✓
Reputation → CoordMedian	PASS	LOW	0.423	✓
<i>C1-FAIL: no constituent suppresses (predicted HIGH)</i>				
Reputation → RFA	C1-FAIL	HIGH	0.825	✓
NormClip → FoolsGold [†]	C1-FAIL	HIGH	0.887	✓
RFA → FoolsGold [†]	C1-FAIL	HIGH	0.914	✓
<i>C2-FAIL: discriminative property destroyed (predicted HIGH)</i>				
NormClip → Reputation	C2-FAIL	HIGH	0.819	✓
<i>C3-FAIL: margin compressed (predicted HIGH)</i>				
FoolsGold → TrimmedMean	C3-FAIL	HIGH	0.508	✓
NormClip → RFA	C3-FAIL	HIGH	0.853	✓
<i>False negatives (predicted HIGH, measured LOW)</i>				
NormClip → CM / RFA → CM	C3-FAIL	HIGH	0.433 / 0.485	×

[†]Reclassified from C2-FAIL: by Proposition 1(a) these preserve C2.

1327 committed pixel 0.045 ± 0.013 ; Neurotoxin [Zhang et al., 2022] top-20% at scale 2.0 0.048 ± 0.009 ;
1328 projection only, no decorrelation, $\epsilon=2$: 0.050 ± 0.008 . With joint constraints the sweep over ϵ gives
1329 0.036 ± 0.015 ($\epsilon=0.5$), **0.167 ± 0.069** ($\epsilon=1$), 0.130 ± 0.030 ($\epsilon=2$), 0.124 ± 0.025 ($\epsilon=4$). ASR
1330 peaks at $\epsilon=1$: tighter projection over-constrains the attack, larger ϵ fails to evade RFA, and without
1331 decorrelation FoolsGold suppresses it outright, so the competing constraints leave a narrow effective
1332 window. This is an empirical finding (Theorem 1 concerns mechanism preservation and does
1333 not bound an adaptive adversary whose optimization is untied to its conditions) and it probes one
1334 natural family of joint constraints at a handful of ϵ values, not the adaptive attack space. Within it,
1335 0.167 is $3.7\times$ the seed-matched committed-pixel ASR ($2.6\times$ the $n=30$ pixel mean of 0.064 ; against
1336 the pairs scaling arm, which the adaptive family does not attack, the ratios would read $3.3\times$ and
1337 $1.8\times$).

1338 **Scale.** At $N=100$, $f=0.05$, ResNet18 both pre-registered PASS predictions hold: FG→RFA max
 1339 ASR = 0.069, FG→CM = 0.152. We flag that these runs reach only ≈ 0.29 clean accuracy at 50
 1340 rounds (ResNet18 is far from converged at this budget), so they establish that suppression transfers
 1341 across scale and architecture, not that the composition is deployable at $N=100$ as configured.

1342 O.1 Heterogeneity, and where the criterion breaks

1343 Earlier drafts swept heterogeneity for single defenses only; no composition had been run at any
 1344 $\alpha \neq 0.5$. We therefore ran all three PASS pairs across $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both attacks,
 1345 50 rounds; max-committed ASR / clean accuracy):

	Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
1346	FG→RFA	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
	FG→CM	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
	Rep→CM	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

1347 In our sweep, which is three seeds per cell and fixes direction rather than magnitude, the certified
 1348 compositions stay below threshold for $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID setting where one
 1349 might expect FoolsGold to founder on legitimately similar benign updates: $\alpha=10$ is indistinguish-
 1350 able from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$) two of the three PASS compositions**
 1351 **cross the threshold** (FG→RFA 0.569, Rep→CM 0.548) and only FG→CM survives, at 0.433: a
 1352 real boundary condition on the framework, with caveats in both directions: FG→RFA at $\alpha=0.1$ is
 1353 strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is
 1354 not, and clean accuracy degrades badly (0.20–0.48), so the regime stresses utility as well as secu-
 1355 rity. Crucially Theorem 1 supplies a diagnostic quantity that retrospectively explains this boundary:
 1356 re-measuring the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$
 1357 collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%), over the
 1358 payload-carrying rounds alone, the ones Figure 6 plots, the same collapse reads 4.71 ± 1.51 ($n=17$)
 1359 to 2.06 ± 2.35 ($n=12$) with the per-round minimum falling $2.82 \rightarrow 0.00$, the difference being that
 1360 the all-round means average in the Lemma 1 rounds at $m=0$, exactly the boundary $m \rightarrow 1$ where
 1361 Corollary 1’s bound degrades to 40% adversarial mass, because extreme heterogeneity inflates the
 1362 benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which
 1363 is the precondition the theorem requires. A practitioner running the 5–10 round pilot we recommend
 1364 would measure $m \approx 1$ and decline to deploy. The criterion is therefore mechanistically diagnostic at
 1365 this boundary (the property that makes a screening rule usable) though this is a post-hoc explanation
 1366 of a known failure, not a pre-registered prediction of it.

1367 O.2 Composition vs. randomized switching

1368 **High persistence sharply reduces the value of temporal switching.** In our tested regime,
 1369 $\gamma \approx 0.66$ ($\gamma := \text{ASR}(t+1)/\text{ASR}(t)$ after one clean round): triggers decay slowly enough that
 1370 intermittent suppression cannot clear them, so temporal mixing approaches the weaker constituent
 1371 defense’s performance. Whether switching retains any value depends jointly on γ , injection strength,
 1372 switching frequency, and recovery dynamics: persistence alone does not imply switching is worth-
 1373 less. The quantitative boundary is architecture-specific; the qualitative prediction (high- γ attacks
 1374 blunt switching) is what this secondary analysis contributes, and nothing in the preservation argu-
 1375 ment rests on it.

1376 **Order-sensitivity control.** Both variants apply both defenses every round and differ *only* in order,
 1377 so compute is matched and any gap is structural. We report it as a **motivating demonstration**
 1378 **that composition order is structurally consequential, not as evidence for the criterion:** the
 1379 mechanism behind the largest gap is a degeneracy rather than a signal-preservation failure, as the
 1380 diagnosis below makes clear. Pixel ASR, 5 seeds: fixed FG→CM 0.076 ± 0.035 versus randomized
 1381 50/50 ordering 0.681 ± 0.133 ($9\times$) and temporal mixing of one defense per round 0.679 ± 0.159 ;
 1382 fixed FG→RFA 0.037 ± 0.011 versus randomized 0.719 ± 0.113 ($19\times$). The mechanism is the
 1383 degeneracy of §1: CoordMedian supplies no per-client transform, so every CM→FG round reduces
 1384 to FoolsGold alone, leaving half the rounds defended by a single constituent that cannot suppress
 1385 on its own. On a reputation+TrimmedMean menu ($n=30$), composition reaches ASR 0.316 against
 1386 0.915 for temporal mixing.

1387 P The metric-swap suite’s two-sided prediction

1388 Because C1 binds for all eight pairs, every frozen label in the metric-swap suite was HIGH (a
 1389 consequence of the measured inputs rather than a later choice), so the suite bears on specificity only.
 1390 One prediction inside it was genuinely two-sided. `cos_krum`’s C1 verdict on model-scaling turns
 1391 on whether the accuracy gate is applied (0.078 ASR at 0.146 clean accuracy), so we froze *both*
 1392 readings in advance: primary (gated) $\Rightarrow$ HIGH, secondary (numeric only) $\Rightarrow$ LOW, with the rule
 1393 fixed beforehand that a LOW outcome at accuracy ≥ 0.35 would mean the gate was wrong. Both
 1394 cells came out **HIGH** (0.508, 0.566), so the gate is vindicated against a pre-registered alternative
 1395 that would have overturned it. This is the only place in the paper where our own accuracy gate was
 1396 put at risk by a frozen rule.

1397 Q The prospective suite, arm by arm

1398 (i) **FLTrust as d_1 (7 pairs, 2/7): out of scope.** When d_1 alone suppresses, the composition in-
 1399 herits that suppression whatever d_2 does, so all five predicted-HIGH pairs are misses even though
 1400 their C2/C3 failures are real: FLTrust genuinely destroys Reputation’s consensus-distance signal,
 1401 but destroying a mechanism that was never load-bearing costs nothing. This is C0, and it was invis-
 1402 ible on the original 42 pairs, where the strongest single defense reaches only 0.519 and the lowest
 1403 ASR among all 39 non-PASS pairs is 0.433; no pair was ever out of scope. We apply C0 here
 1404 *retrospectively*, which is weaker than a pre-registered exclusion.

1405 (ii) **FLTrust as d_2 (6 pairs, 6/6): in scope, but nearly an identity.** Proposition 1(a) predicts
 1406 $\cdot \rightarrow$ FLTrust preserves C2, and it does, but FLTrust gates on a scale-invariant cosine and then *renor-*
 1407 *malizes every update to* $\|g_{\text{srv}}\|$, discarding magnitude outright, so a positive rescaling $u_i \mapsto c_i u_i$
 1408 is *exactly* cancelled: FLTrust discards precisely what d_1 modifies. The per-seed data show this di-
 1409 rectly. $\text{Krum} \rightarrow$ FLTrust and $\text{MKrum} \rightarrow$ FLTrust are **bit-identical** to FLTrust alone on all six runs
 1410 (pass-through d_1); $\text{NC} \rightarrow$ FLTrust is bit-identical *on the pixel attack*, where no client exceeds $\tau=5$
 1411 so $c_i \equiv 1$, and differs only within seed noise on model-scaling, where clipping fires; $\text{Rep} \rightarrow$ and
 1412 $\text{RFA} \rightarrow$ FLTrust likewise differ only within noise. The lone real deviation is $\text{FG} \rightarrow$ FLTrust (0.064,
 1413 up to 0.084 per seed): the one d_1 producing exact *zeros* rather than positive rescalings, changing
 1414 FLTrust’s client set instead of its magnitudes, which is the Lemma 1 interaction. So this arm con-
 1415 firms that theory and implementation agree, and demonstrates the degeneracy mechanism cleanly,
 1416 but it is **not falsifiable**: every d_1 in our menu is a *non-negative* per-client rescaling, FLTrust cancels
 1417 the positive part exactly, and so the only channel left open is the *zero pattern*, which is precisely
 1418 where the one deviation appears. Proposition 1(a) permits no other outcome on the strictly-positive
 1419 part, and Lemma 1 accounts for the rest.

1420 (iii) **Neither defense is FLTrust (7 pairs, 5/7): the only arm that could have surprised us.**
 1421 All were predicted HIGH; correct are $\text{rep} \rightarrow \text{krum}$ 0.880, $\text{nc} \rightarrow \text{mkrum}$ 0.943, $\text{rfa} \rightarrow \text{mkrum}$ 0.875,
 1422 $\text{krum} \rightarrow \text{rfa}$ and $\text{mkrum} \rightarrow \text{rfa}$ both 0.870. Of those five, only $\text{rep} \rightarrow \text{krum}$ composes two real mecha-
 1423 nisms: MultiKrum at $K=5$ is bit-identical to FedAvg, so $\text{nc} \rightarrow \text{mkrum}$ and $\text{rfa} \rightarrow \text{mkrum}$ are Norm-
 1424 Clip and RFA standalone, and $\text{krum} \rightarrow \text{rfa}$ and $\text{mkrum} \rightarrow \text{rfa}$ are one measurement of RFA standalone
 1425 made twice (per-seed $|\Delta \text{ASR}| = 0.000$). Neither miss is a mechanism error: $\text{fg} \rightarrow \text{krum}$ reads 0.078
 1426 at 9.9% clean accuracy (a collapsed model, the same hollow-suppression artifact as FoolsGold alone)
 1427 and $\text{krum} \rightarrow \text{cm}$ reads 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian
 1428 alone, whose own baseline is 0.519: threshold noise on a borderline constituent. This arm contains
 1429 *zero* predicted-LOW pairs, so like Tier 2 it bears on specificity only.

1430 **The suggestive contrasts, withdrawn.** We had read three matched pairs, d_1 held fixed while d_2
 1431 moved from FLTrust to a Euclidean rule, as a 19–20 $\times$ gap turning on exactly the property Propo-
 1432 sition 1 identifies. Two of them ran against MultiKrum, which at $K=5$ selects *all five* participants
 1433 and is therefore bit-identical to FedAvg (1.9×10^{-6} per aggregation; a genuine selector again at
 1434 $K=10$), so those arms carried no downstream defense at all. Substituting the genuine selector Krum
 1435 does not repair them, and that is the finding: all three Krum arms take their max-committed ASR
 1436 from the pixel backdoor, on which Krum’s own standalone ASR is 0.583, so the gap measures C1
 1437 failure rather than disturbed invariance, and $\text{nc} \rightarrow \text{krum}$ is bit-identical to Krum alone on all three
 1438 seeds because under that attack no client exceeds $\tau=5$ and d_1 is the exact identity. Counting the
 1439 pass-through d_1 rows as well, 8 of the 20 rows reduce to a single defense rather than the 5 labelled

DEGEN at freeze, and arm (iii) composes two real mechanisms in 2 pairs, not 7. The frozen labels and the 13/20 score stand; what we retract is our reading of them. The confound is accordingly worse than we reported, in the way Lemma 2 predicts: the invariant-signal d_2 is also the only d_2 in this menu that suppresses alone, so effectiveness and mechanism co-vary across the contrast. This is why breaking the confound requires holding standalone effectiveness fixed rather than choosing a different defense.

R Adversary-correlated dispersion Δ_c , measured

R.1 Heterogeneity is not the mechanism: ρ against Δ_c

The finding, stated first. What costs a downstream defense its suppression is not how heterogeneous the upstream transform is but **whether its reweighting is correlated with adversary identity**, and not even the size of that correlation, which has **two opposite regimes**. This is why our own headline predictor failed.

Two different things a coefficient vector does, and the theory sees only one. (Symbols: Appendix G.) Proposition 1 turns on **dispersion**, $\rho := \max_i c_i / \min_i c_i$ (how far apart the coefficients are, which decides whether an ordering survives) and says nothing about *which* clients receive which coefficient. That second property is **adversary-correlated dispersion**,

$$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A], \quad A \text{ the adversarial participants,}$$

and the two are **independently controllable**, not merely uncorrelated in our sample: a transform can spread the coefficients arbitrarily at $\Delta_c = 0$ exactly, or leave them nearly equal while shifting the adversary’s share, and Proposition 3 exhibits a family of each. **Proposition 1 and Theorem 1 constrain ρ ; Lemma 1’s attenuation channel is about Δ_c** , and §5 finds the real transforms differ on the second. Δ_c is *not* a screening quantity: computing it requires knowing A (§6).

Two mechanism-preserving regimes, at opposite ends of ρ . Lemma 1 is a second, independent route to suppression, and the two together explain all three PASS compositions. Let $r := w_a / \max_b w_b$ and S be the attack’s coordinate extremeness. **(A) Ordering preservation** keeps the adversary in the tail where a rank aggregator discards it ($S \cdot r > 1$). **(B) Attenuation** instead drives the adversarial transformed value *inside* the benign spread, where it cannot be an extreme order statistic and its payload is scaled by r ($S \cdot r \ll 1$; $r = 0$ is exact annihilation). The hazard is $S \cdot r \approx 1$, and our two weighting defenses sit at opposite ends: FoolsGold’s weights are bounded ($\rho \leq 2.1$, $S \cdot r \geq 4.8$), regime A; Reputation’s are wildly unbounded (ρ mean 1.4×10^5) precisely *because* it crushes the adversary ($S \cdot r = 0.017$), regime B. So Reputation $\rightarrow$ CoordMedian violates the ordering condition by five orders of magnitude and still suppresses (Figure 3, App. N, exhibits both regimes). Hence ρ does not order Δ_c (Proposition 3), and Δ_c ’s *magnitude* does not order suppression either, the same large $|\Delta_c|$ being protective at one end and catastrophic at the other. **Neither quantity is a predictor**; they buy an account of *why* a heterogeneity-based reading mispredicts.

$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ for every per-client transform in the paper, over the same 5 seeds $\times$ 3 rounds and, within a round, the *same* raw update stack for every transform, so the rows are comparable by construction (measure_coefficient_targeting.py, results/coefficient_targeting.json). **No ASR is computed and no model is trained per transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed norms, $c_i = \|T(u_i)\|/\|u_i\|$, so Table 8 measures the shipped code path rather than the builder’s description of it. As a provenance check the synthetic families’ adversarial coefficient shares reproduce results/admission_measurement.json to 0.0 absolute on all 13 cells, so these rows may be read on the same axis as the Mode-S and Mode-A figures of §4.

Three readings, in decreasing confidence. **(1) ρ is not a proxy for Δ_c** : the two are anti-ordered on the anchors, and NormClip on the pixel backdoor has $\rho = 1$ exactly because no client exceeds $\tau=5$, so $\Delta_c = 0$ there for a reason the invariance classification cannot see. **(2) Both transforms carrying a missed anchor are targeted far beyond any ladder rung**, which puts the residuals of §5 on a measured axis rather than a range mismatch; Δ_c does not sort the anchors themselves,

Table 8: ρ says how much a transform spreads the coefficients; Δ_c says who receives the spread. They are not ordered together: RFA and NormClip target the adversary 2–3.6 $\times$ harder than the ladder’s most dispersed rung at ρ that is 3.7–13 $\times$ *smaller*. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

transform	attack	ρ	$\mathbb{E}[c \mid A]$	$\mathbb{E}[c \mid \neg A]$	Δ_c
NormClip	model-scaling	4.19	0.268	0.989	−0.721
NormClip	pixel	1.00	1.000	1.000	0.000
Reputation	model-scaling	8.3×10^5	0.003	1.387	−1.383
Reputation	pixel	1.68	0.915	1.019	−0.105
FoolsGold	model-scaling	1.9×10^{12}	1.497	0.834	+0.663
FoolsGold	pixel	1.8×10^{12}	1.365	0.986	+0.378
RFA	model-scaling	14.8	0.125	1.332	−1.208
RFA	pixel	1.69	0.923	1.012	−0.089
<i>Synthetic families of §5–§4, same code path:</i>					
ladder, $\kappa=0.5/1.0/2.0$	—	2.72/7.39/54.6	—	—	−0.140/−0.240/−0.336
Mode S, all rungs	—	1.00–54.6	—	—	0.000
Mode A, $\nu=-2 \dots +2$	—	7.39–7.39	—	—	−1.138 $\dots$ + 2.469

1490 since NormClip carries one that misses and one that lands. **(3) Δ_c is nonetheless not a predictor of**
1491 **suppression:** RFA is strongly negative at −1.208 and rfa→krum fails at 0.779, while Reputation is
1492 more negative still at −1.383 and rep→coord_median suppresses. That is the two-regime structure
1493 of §2 again: strong negative Δ_c can mean either annihilation or mere dilution, and Δ_c ’s magnitude
1494 alone does not say which. We report it as the axis on which the ladder’s anchors were unmatched,
1495 not as the quantity that replaces the statistic.

1496 S Confusion matrix in full

1497 Exactly 5 of the 42 pairs have max-committed ASR < 0.5: the 3 PASS pairs plus the 2 false
1498 negatives, all of which appear among the representative rows of Table 7. Pooling both in-distribution
1499 tiers (each pair scored against the label committed for it at the time) gives TP = 3, FP = 2, TN = 35,
1500 FN = 2 (38/42; specificity 94.6%, which we report but do not lead with, since 37 of 42 pairs are
1501 negatives and specificity is inflated by that imbalance; precision 1.0 at base rate 11.9% is the 8.4 $\times$
1502 enrichment that matters, and recall is 40% under the tightened C1, since FG→RFA now counts as a
1503 miss). We report this pooled figure with its decomposition rather than on its own, because the tiers
1504 are not equivalent evidence: it is 16/18 development *fit* plus 22/24 held-out *prediction*. The four
1505 sub-tiers the body refuses to pool are development *fit*, held-out *specificity*, out-of-sample *positives*
1506 *with no informative negatives*, and the 13/20 *prospective* suite. Scoring instead against the criterion’s
1507 mechanism labels, which classify both failed LOW calls as HIGH, gives TP = 3, FP = 0, TN = 37,
1508 FN = 2 (40/42); we take the more conservative pooled figure as primary. The two false positives,
1509 fg→NC (0.927) and fedavg→CM (0.519), were hedged as speculative at pre-registration time and
1510 neither satisfies C1; every pair that actually satisfies C1 $\wedge$ C2 $\wedge$ C3 came out LOW. Category tally
1511 after the Proposition 1 correction: degenerate 18, C1-fail 11, C2-fail 3, C3-fail 7, PASS 3, and all 3
1512 surviving C2 failures are $\cdot \rightarrow$ Reputation, exactly the non-invariant class of Proposition 1(c). With
1513 a positive set of only 2 certified in-distribution pairs and recall 40%, the 94.6% specificity figure
1514 is high partly because the positive class is rare, so we report it as a supporting statistic rather than
1515 a headline. The conditions are sufficient, not necessary, and they do reject working compositions.
1516 Extension to 3-way compositions: 9/10 triples correct (too small for confidence).

1517 T Diagnosing the two false negatives in full

1518 NormClip→CM (0.433) and RFA→CM (0.485) were labelled C3-FAIL on the reading that *any* het-
1519 erogeneous upstream rescaling compresses a rank aggregator’s margin. Theorem 1(1) says some-
1520 thing weaker and quantitative: ordering survives when the attack’s coordinate extremeness exceeds
1521 the realized weight ratio, $S > \rho$. Measured, NormClip’s ratio is mild. Over 9 live rounds it clips
1522 only 0–2 of 5 clients (11 of 45 client-rounds), so $\approx 76\%$ of clients receive $c_i = 1$ *exactly*, and the
1523 largest raw norm we observe is 24.5 against $\tau=5$, bounding $\rho \leq 4.9$; against $S=10$ the margin

1524 $S/\rho \geq 2.0$ still clears 1. RFA-reweighting is likewise near-uniform on benign clients. Both pairs
 1525 therefore sit close to the *uniform* rescaling case, where ordering is preserved exactly, which is why
 1526 their ASR lands in an intermediate band just under threshold rather than in the high band the binary
 1527 label predicts. **The refinement does not recover them.** Both also fail C1: CoordMedian alone is
 1528 already at 0.519, and neither NormClip (0.935) nor RFA (0.885) suppresses, so no reading of C3
 1529 makes either pair certifiable. What the diagnosis buys is a sharper condition and an account of the
 1530 residual: a d_2 whose own baseline sits at 0.519 needs only a mild order-preserving upstream trans-
 1531 form to cross 0.5, so these two are threshold noise around a marginal aggregator, not suppression
 1532 the framework failed to recognise. The operational refinement is to evaluate C3 for rank aggrega-
 1533 tors against the *realized* ρ from a 5–10 round pilot rather than the binary question of whether d_1 is
 1534 heterogeneous: the same quantity Theorem 1(1) already requires, which the labelling had not been
 1535 using.

1536 U Condition-by-condition scoring, and the run-count saving

1537 The results below are §3’s, moved here for space. None changes a reported number.

1538 **The out-of-sample bookkeeping, in full.** All 4 out-of-sample PASS predictions the tightened cri-
 1539 terion licenses held, as did both in-distribution $C1 \wedge C2 \wedge C3$ pairs. **But all 7 are *positive* predictions**
 1540 **whose FAIL controls are uninformative** (DBA fails even undefended), which is why §3 declines
 1541 to read them as an out-of-sample classification result: five positive predictions against an 11.9%
 1542 base rate (5/42, Appendix S), with the positive direction separately unvalidated (§4).

1543 **Does every condition earn its place? C3 does not, reported as a finding.** Scoring all 42 pairs
 1544 on each condition *independently*: C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, $C1 \wedge C2$
 1545 selects 2 at 100%, and **C3 alone selects 35 at 9%**, below the 12% base rate, never binding on a
 1546 pair $C1 \wedge C2$ already admits. So **we demote C3 to a margin diagnostic and recommend $C1 \wedge C2$,**
 1547 retaining it on theoretical grounds only: Theorem 1 makes margin compression a real failure mode
 1548 and the $\alpha=0.1$ failure of §3 is exactly a margin collapse. It changes no number: a finding about the
 1549 criterion, not an improvement to it. **One scope limit on the C2 column that we describe as read**
 1550 **off the two defenses’ definitions:** for the conditional classes (*coord_median*, *trimmed_mean*, *rfa*)
 1551 it is not, and *c2_provenance* in the same generator censuses the difference: 3 of the 14 pairs C2
 1552 admits rest on algebraic invariance and 11 on a per-pair call made in sample, those 11 containing all
 1553 5 low-ASR pairs. *Within this menu* the algebraically settled region holds 3 pairs and no positive, so
 1554 here C2’s own coverage is carried by judgement, not by Prop. 1. **That second half we then removed**
 1555 **by enlarging the region rather than relabelling it:** those 3 pairs are one column of a 3×3 census,
 1556 and closing it prospectively puts a low-ASR pair inside the algebraic region and separates the two
 1557 strategies more sharply there than anywhere in this table (§K.2). What is disclosed and not repaired
 1558 is the *first* half, which needs an invariance result for the conditional classes; we do not have one.

1559 **What the saving is.** Counted in runs rather than pairs (C1’s standalone evaluations are runs too)
 1560 the saving is 420 **unscreened against 80 screened**, 81% **fewer composition-evaluation runs**, not
 1561 the 95.2% of *pairs* we previously reported; **nothing is eliminated before any run**, and the structural
 1562 claim is the scaling $|A| S D(D-1)$ (*quadratic* in the defense count) to $|A| S ((D-1) + |\text{admitted}|)$.
 1563 GPU-hour figures are *estimates*; no timing telemetry was recorded.

1564 V What the boundary permits, claim by claim

1565 The inventory below separates the three kinds of *no* this paper reports: *structural* (no design satis-
 1566 fying the gate can answer the question, Prop. 4), *open* (answerable in principle, but not by an oracle
 1567 instrument), and *scope* (outside what the committed-attack setting measures). Nothing here is a new
 1568 claim; each row points at the section that establishes it.

1569 W Anticipated objections

1570 Five objections we expect, answered against text that is already in the paper rather than by new
 1571 argument. Nothing here is a new claim; each answer is a pointer plus the scope it turns on.

Table 9: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 4). Every entry is scoped to the measured setting of §5.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: unchanged in every one of the 60 measured rounds
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside margin (Prop. 2(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes, non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No: refuted: +0.098 against a frozen +0.150
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos_krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no: open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no: structural	Empty cell by the gate, for every menu (Prop. 4)
Does any of this certify <i>deployed</i> security?	no: scope	Committed attack only; one adaptive test breaks it

1572 **1. This is small-scale federated learning, not decentralized foundation-model training.**
1573 Agreed, and we claim **methodological** relevance only (§1): no transformer arm, no foundation-
1574 model measurement, and no sentence implying either. What transfers is a statement about *causal*
1575 *structure*: any criterion whose eligibility gate is defined on the same outcome it predicts conditions
1576 on a descendant of both mechanism and effectiveness (Prop. 4), and that argument does not depend
1577 on model scale or participant count. What does *not* transfer is every empirical magnitude, which
1578 is why each is reported per arm, tiered at **very unequal weight and never pooled** (Table 2), and
1579 scoped in Table 1. The result forbids an inference; it does not measure a system.

1580 **2. The interventions are oracles: no deployed system can pin an adversary’s coefficient or**
1581 **read adversary identity.** Correct, and that is why Mode S is an **instrument, not a screen**. The
1582 Conclusion says so in those terms: Mode S measures level (P4) only by reading adversary identity,
1583 which no deployable defense can, so our experiments **identify the causal channel and are not a**
1584 **(P4) screen**. Pinning $c=1.0$ has one purpose, to cut the confounded path of Figure 1(a) by construc-
1585 tion, exactly as an oracle-labelled control isolates a channel in an ablation. The deployable question
1586 is left open as the paper’s closing sentence. The one control that *removes* an oracle component
1587 (score-only, which scores the transformed stack but aggregates the untransformed selected update)
1588 does not change the verdict (App. E, D).

1589 **3. $n=5$ seeds is small.** It is: $n=5$ per arm, $n=8$ for `cos_krum`, with Student- t 95% intervals
1590 throughout and no claim that a small effect is zero. Where suppression does not move we report **no**
1591 **evidence of a practically meaningful change under the pre-specified ± 0.15 equivalence margin**,
1592 a margin frozen before the runs (Table 1), never a claim that suppression is unchanged. Two design
1593 choices do more for precision than seeds would: the identity rungs are imported **bit-identical per**
1594 **seed** (largest deviation 0.00e+00), so the contrast is within-seed rather than across noise, and the
1595 central claim is a **sign reversal** (-0.272 against $+0.098$ on the identical cell, Table 3), which is not
1596 a null result. Finally, Prop. 4 says the cell that would validate the criterion is *empty by design*: more
1597 seeds cannot repair it (Table 9).

1598 **4. If ASR does not move, why does a change of decision matter?** Because the change of de-
1599 cision is what a preservation criterion *measures*, and the gap between the two is the finding. An
1600 invariance argument certifies the top three levels of Def. 1 while the security content sits at level
1601 (P4), the adversarial contribution admitted. A criterion reading the Krum arm would call that com-
1602 position broken (the decision flips in up to 80% of rounds) for a composition whose suppression
1603 shows no measurable change: a false alarm in the direction that costs coverage (Prop. 5 caps such a
1604 screen’s recall at 80%). The converse fails too: `cos_krum`’s selection is bit-identical at every rung
1605 yet it never suppresses (0.584 at the identity) and its ASR *falls* 0.173 at the first accuracy-gate-
1606 passing rung, the full-ladder endpoint of 0.425 lying below the accuracy floor, so exact invariance
1607 of the statistic neither implies suppression nor holds it fixed (App. J.5).

1608 **5. The theory covers only positive per-client rescalings.** Yes, and we say so wherever it is used:
1609 **a laboratory for studying composition, not a general model of it** (§1; Prop. 1, Thm. 1, App. I).
1610 The class is not arbitrary (it is what norm clipping, contribution weighting and reputation weighting
1611 do to a client’s update) but it excludes coordinate-wise filters and any transform that changes an

1612 update’s direction, and the theorem’s own margin closes under heterogeneity ($m=3.24 \rightarrow 1.01$ at
 1613 $\alpha=0.1$), so **it certifies nothing where heterogeneity is high**. The negative does not rest on the
 1614 theory: it is a *measured* counterexample, and a narrower theory makes the counterexample sharper
 1615 rather than weaker, since **the theorem certifies mechanism preservation, not security**, it bounds
 1616 no ASR.

1617 X Experimental Details

1618 Figure 7 places the evaluated settings in the (γ, p_{eff}) plane. At the persistence we measure ($\hat{\gamma} \approx$
 1619 0.66) the one temporal mix we ran reaches ASR 0.956, which is the reason the paper composes
 1620 rather than switches; the figure records those points and draws no boundary between them.

1621 **FL configuration.** $N=10$ clients, $K=5$ per round, 50 rounds, SGD with momentum 0.9, learning
 1622 rate 0.01, local epochs 3, batch size 32, Dirichlet $\alpha=0.5$ (non-IID). Scale experiments: $N=100$,
 1623 $K=10$, $f=0.05$, ResNet18 with GroupNorm.

1624 **Attacks.** Model-scaling: scale factor 10, 50% poison fraction, target class 0. Pixel backdoor:
 1625 5×5 bottom-right trigger, target class 0. DBA: 4 sub-triggers at image corners, $f=0.4/0.6$. Joint-
 1626 constraint: $\epsilon \in \{0.5, 1.0, 2.0, 4.0\}$, disjoint coordinate blocks per adversary, single-step ϵ -ball pro-
 1627 jection per round.

1628 **Defenses.** NormClip $\tau=5.0$, i.e. $u_i \mapsto u_i \min(1, \tau/\|u_i\|_2)$ over the whole flattened update;
 1629 TrimmedMean $\beta=0.2$ (trim 20% each tail); RFA: unweighted Weiszfeld, 50 iterations, tolerance
 1630 10^{-6} , initialised at the mean; FoolsGold: $w_i \propto \log((1 - s_i)/(s_i + \epsilon) + \epsilon)$ with s_i the client’s
 1631 maximum absolute pairwise cosine similarity normalised by the round maximum, $\epsilon=10^{-5}$, then
 1632 shifted by $-\min_i$ and normalised, so the least-trusted client of each round receives weight exactly
 1633 0 (empirically 2 of 5 clients per round).

1634 **Reputation.** Despite the name this defense is *memoryless*, it keeps no cumulative per-client state
 1635 and is recomputed from scratch each round from that round’s K participants. Consensus is the
 1636 coordinate-wise median m of the K updates; scores are ℓ_2 distances $d_i = \|u_i - m\|_2$; the bandwidth
 1637 is adaptive, $\sigma = \text{median}_i(d_i)$ (clamped $\geq 10^{-6}$); weights are $w_i = \exp(-d_i/\sigma)$, normalised to sum

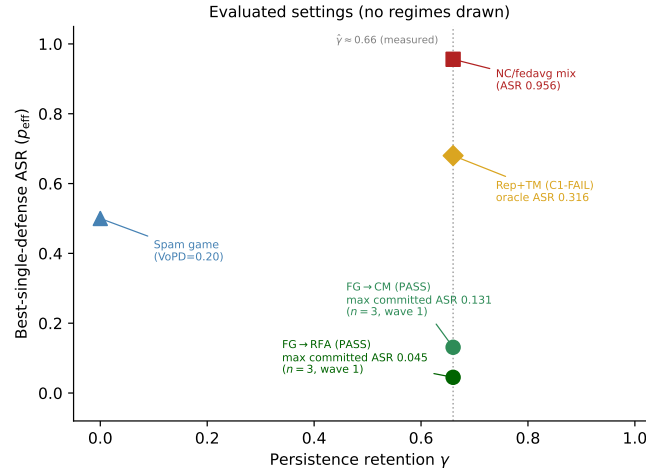


Figure 7: **The evaluated settings, plotted. No regimes, and one line, which is a measurement.** X-axis: persistence γ . Y-axis: best single-defense ASR (p_{eff}). Each marker is a setting we ran and carries its own measured value; the dotted line is the measured FL persistence $\hat{\gamma} \approx 0.66$. **An earlier version shaded four regime bands and drew threshold guides at $\gamma=0.15$, $p_{\text{eff}}=0.3$ and $p_{\text{eff}}=0.6$; none of those numbers was measured, so all six are gone.** Five points cannot locate a boundary, and nothing here is evidence of a phase transition in (γ, p_{eff}) space.

1638 to 1. No clipping, no initialisation. Its discriminative property is therefore *consensus distance*, not
 1639 similarity spread, which is why Proposition 1(c) predicts that $\cdot \rightarrow$ Reputation compositions fail C2.
 1640 Weights are strictly positive, so Reputation never produces the zero vectors of Lemma 1.

1641 **Adversary count per round.** Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$; each
 1642 round samples K participants uniformly without replacement from all N , with no stratifica-
 1643 tion. For $N=10$, $f=0.2$, $K=5$ the per-round adversary count is hypergeometric: $P(n_a=0) =$
 1644 $P(n_a=2) = 56/252 = 0.222$, $P(n_a=1) = 140/252 = 0.556$; measured over 100 rounds we ob-
 1645 serve 0.19/0.54/0.27. Two consequences, both matching the data. CoordMedian’s count majority
 1646 $n_a < K/2 = 2.5$ holds for every attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds. TrimmedMean’s
 1647 budget is $\lfloor \beta K \rfloor = \lfloor 1.0 \rfloor = 1$, so it is exceeded whenever $n_a=2$: 22.2% of rounds, ≈ 11 of 50.
 1648 This is why $FG \rightarrow$ TrimmedMean fails (0.508) while $FG \rightarrow$ CoordMedian succeeds (0.107) under an
 1649 otherwise identical upstream transform. We retain the pre-registered C3-FAIL label for $FG \rightarrow$ TM;
 1650 the count-budget violation is an additional mechanism, not a relabelling.

1651 **Theorem quantities.** FoolsGold weight ratio across 5 seeds $\times$ 20 rounds ($N=10$, $f=0.2$, model-
 1652 scaling): $\max \rho = 2.1$, giving extremeness ratio $S/\rho \geq 4.8$ with $S=10$ (regime A). Reputation,
 1653 same protocol (5 seeds $\times$ 20 rounds, 81 rounds containing both adversarial and benign clients): ρ
 1654 mean 1.44×10^5 , max 2.84×10^6 , p95 6.1×10^5 ; mean adversarial weight 4.83×10^{-4} against mean
 1655 benign weight 0.278, and $w_a < 10^{-3}$ in 85% of rounds. Reputation therefore fails the ordering
 1656 condition by ~ 5 orders of magnitude and suppresses via regime B instead. RFA, measured on
 1657 the transformed points $u'_i = w_i u_i$ that RFA actually receives (25 adversary-present rounds over 3
 1658 seeds): FG zeroes every adversary in 32% of rounds, where Lemma 1 applies and the measured
 1659 contraction $1 - \Lambda_a$ is 0.29 on average (range 0.001–0.52). In the other 68% ($w_a > 0$) the separation
 1660 condition of Theorem 1(2) is satisfied in *every* round, with margin $\Delta'_{\text{sep}}/2(R'_B + \delta') \geq 2.8$ (mean
 1661 3.2) and δ'/R'_B mean 0.19, max 0.72. The two cases are exhaustive, so the analysis covers all
 1662 measured rounds. Proposition 1(a) verified in the shipped implementation: FoolsGold’s weight
 1663 vector changes by at most 3×10^{-8} (float32 rounding) under NormClip across 1000 synthetic trials
 1664 and live rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against
 1665 $\tau=5$). Exactly 2 of the 5 clients per round receive zero FoolsGold weight. **Corollary 1 in full:**
 1666 with $D_b := R'_B + \delta'$ bounding every benign residual and $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$
 1667 lower-bounding every adversarial one, the Weiszfeld coefficients satisfy $\sum_j 1/d_j \geq n_b/D_b$ and
 1668 $1/d_a \leq 1/D_a$, so $\Lambda_a \leq (n_a/D_a)/(n_b/D_b + n_a/D_a) = n_a/(n_b(2m - 1) + n_a)$. At $m \rightarrow 1$ this
 1669 is $n_a/(n_b + n_a)$, i.e. 40% for $n_a=2$, $n_b=3$; at the measured $m \geq 2.82$ it is $\leq 12.6\%$, and at the
 1670 all-round mean $m = 3.24$, $\leq 10.8\%$.

1671 **Pre-registration provenance.** We distinguish the tiers explicitly. The 18-pair development set is
 1672 where the criterion was *developed*; its predictions were not committed in advance and we report it
 1673 as fit. The 24-pair held-out set was pre-registered in `pre_registration_wave2.md` and committed
 1674 before those experiments ran; it contains no predicted-PASS pair, so it bears only on specificity. The
 1675 out-of-sample PASS predictions (DBA transfer, $N=100$ /ResNet18 scale transfer, composition dom-
 1676 inance) were pre-registered separately in `pre_registration_oos_pass.md`, also committed in ad-
 1677 vance. The 10 three-way predictions were pre-registered in `pre_registration_three_way.md`.
 1678 Two held-out LOW calls, hedged as speculative at pre-registration time, failed; we score
 1679 them as misses rather than dropping them. The prospective suite of §5 was pre-registered
 1680 in `pre_registration_prospective.md` at commit 262cf35, which precedes the first write
 1681 to `results/prospective_suite/`; single-defense baselines for the three new defenses were
 1682 measured *before* the freeze, because C1 and C0 take them as inputs (FLTrust 0.048, Krum
 1683 0.583, Multi-Krum 0.783 max-committed; FedAvg 0.667 scaling / 0.768 pixel, clearing the
 1684 sanity gate). The predicted labels in that file are reproduced verbatim in the PAIRS list of
 1685 `run_prospective_suite.py` and the confusion matrix is recomputed from per-seed ASR in
 1686 `results/prospective_suite/summary.json`, not transcribed.

1687 **Freeze commits and run tallies for the interventional suites.** Every design,
 1688 grid and threshold below was git-committed before the corresponding results direc-
 1689 tory existed, and each hash is verifiable in the repository history. *Dose-response*
 1690 *ladder* (§5): `pre_registration_dose_response.md` at commit e711a95, before
 1691 `results/dose_response/`; 80 runs. *Metric-swap mirrors* (§5): 8 frozen labels at commit

1692 356897b. *Prospective suite* (§5): commit 262cf35, as above. *Targeted intervention, Modes S*
 1693 *and A* (§4, §5): design and thresholds at commit 5130cec before results/targeted_dose/
 1694 existed; 97 runs plus 25 identity runs verified bit-identical per seed (largest deviation 0.00e+0);
 1695 the adversarial coefficient share was verified with no ASR involved, before any run. *Rank-*
 1696 *discordant replication* (§4): pre_registration_dose_replication.md at commit 06d9bce
 1697 before results/dose_replication/ existed; 15 new runs, identity rung imported bit-identical
 1698 from the ladder and verified by --harness-check. *Second dataset and architecture* (§4):
 1699 pre_registration_dose_femnist.md at commit 478b555 before results/dose_femnist/
 1700 existed; 12 new runs, with $\kappa=0$ run in-suite because no EMNIST-byclass ladder exists to
 1701 import from. *Score-only control* (§4, Appendix E): pre_registration_score_only.md at
 1702 commit 35788d9 before results/score_only/ existed; 15 new runs, identity rung imported
 1703 bit-identical and verified by --harness-check, and the cos_krum corollary checked at one seed
 1704 as pre-registered.